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FLUID ROUTING PLUG

Abstract

A fluid routing plug for use with a fluid end section. The fluid end section being one of a plurality of fluid end sections making up a fluid end side of a high pressure pump. The fluid routing plug is installed within a horizontal bore formed in a fluid end section and is configured to route fluid throughout the fluid end section.

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Background/Summary

BACKGROUND

[0001] Various industrial applications may require the delivery of high volumes of highly pressurized fluids. For example, hydraulic fracturing (commonly referred to as “fracking”) is a well stimulation technique used in oil and gas production, in which highly pressurized fluid is injected into a cased wellbore. As shown for example in FIG. **1**, the pressured fluid flows through perforations **10** in a casing **12** and creates fractures **14** in deep rock formations **16**. Pressurized fluid is delivered to the casing **12** through a wellhead **18** supported on the ground surface **20**. Sand or other small particles (commonly referred to as “proppants”) are normally delivered with the fluid into the rock formations **16**. The proppants help hold the fractures **14** open after the fluid is withdrawn. The resulting fractures **14** facilitate the extraction of oil, gas, brine, or other fluid trapped within the rock formations **16**.

[0002] Fluid ends are devices used in conjunction with a power source to pressurize the fluid used during hydraulic fracturing operations. A single fracking operation may require the use of two or more fluid ends at one time. For example, six fluid ends **22** are shown operating at a wellsite **24** in FIG. **2**. Each of the fluid ends **22** is attached to a power end **26** in a one-to-one relationship. The power end **26** serves as an engine or motor for the fluid end **22**. Together, the fluid end **22** and power end **26** function as a high pressure pump.

[0003] Continuing with FIG. **2**, a single fluid end **22** and its corresponding power end **26** are typically positioned on a truck bed **28** at the wellsite **24** so that they may be easily moved, as needed. The fluid and proppant mixture to be pressurized is normally held in large tanks **30** at the wellsite **24**. An intake piping system **32** delivers the fluid and proppant mixture from the tanks **30** to each fluid end **22**. A discharge piping system **33** transfers the pressurized fluid from each fluid end **22** to the wellhead **18**, where it is delivered into the casing **12** shown in FIG. **1**.
[0004] Fluid ends operate under notoriously extreme conditions, enduring the same pressures, vibrations, and abrasives that are needed to fracture the deep rock formations **16**, shown in FIG. **1**. Fluid ends may operate at pressures of 5,000-15,000 pounds per square inch (psi) or greater. Fluid used in hydraulic fracturing operations is typically pumped through the fluid end at a pressure of at least 8,000 psi, and more typically between 10,000 and 15,000 psi. However, the pressure may reach up to 22,500 psi. The power end used with the fluid end typically has a power output of at least 2,250 horsepower during hydraulic fracturing operations. A single fluid end typically produces a fluid volume of about 400 gallons, or 10 barrels, per minute during a fracking operation. A single fluid end may operate in flow ranges from 170 to 630 gallons per minute, or approximately 4 to 15 barrels per minute. When a plurality of fluid ends are used together, the fluid ends collectively may deliver as much as 4,200 gallons per minute or 100 barrels per minute to the wellbore.
[0005] In contrast, mud pumps known in the art typically operate at a pressure of less than 8,000 psi. Mud pumps are used to deliver drilling mud to a rotating drill bit within the wellbore during drilling operations. Thus, the drilling mud does not need to have as high of fluid pressure as fracking fluid. A fluid end does not pump drilling mud. A power end used with mud pumps typically has a power output of less than 2,250 horsepower. Mud pumps generally produce a fluid volume of about 150-600 gallons per minute, depending on the size of pump used.
[0006] In further contrast, a fluid jetting pump known in the art typically operates at pressures of 30,000-90,000 psi. Jet pumps are used to deliver a highly concentrated stream of fluid to a desired area. Jet pumps typically deliver fluid through a wand. Fluid ends do not deliver fluid through a wand. Unlike fluid ends, jet pumps are not used in concert with a plurality of other jet pumps. Rather, only a single jet pump is used to pressurize fluid. A power end used with a jet pump typically has a power output of about 1,000 horsepower. Jet pumps generally produce a fluid volume of about 10 gallons per minute.
[0007] High operational pressures may cause a fluid end to expand or crack. Such a structural failure may lead to fluid leakage, which leaves the fluid end unable to produce and maintain adequate fluid pressures. Moreover, if proppants are included in the pressurized fluid, those proppants may cause erosion at weak points within the fluid end, resulting in additional failures.
[0008] It is not uncommon for conventional fluid ends to experience failure after only several hundred operating hours. Yet, a single fracking operation may require as many as fifty (50) hours of fluid end operation. Thus, a traditional fluid end may require replacement after use on as few as two fracking jobs. There is a need in the industry for a fluid end configured to avoid or significantly delay the structures or conditions that cause wear or failures within a fluid end.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
First Embodiment
[0009] FIG. **1** is an illustration of the underground environment of a hydraulic fracturing operation.
[0010] FIG. **2** illustrates above-ground equipment used in a hydraulic fracturing operation.
[0011] FIG. **3** is a front perspective view of one embodiment of a high pressure pump disclosed herein.
[0012] FIG. **4** is a front perspective view of the fluid end assembly shown in FIG. **3** attached to a

plurality of stay rods.
[0013] FIG. **5** is a rear perspective view of the fluid end assembly and stay rods shown in FIG. **4**.
[0014] FIG. **6** is a front elevational view of the fluid end assembly and stay rods shown in FIG. **4**.
[0015] FIG. **7** is a front perspective view of the fluid end assembly shown in FIGS. **3** and **4**.
[0016] FIG. **8** is a rear perspective view of the fluid end assembly shown in FIG. **7**.
[0017] FIG. **9** is a cross-sectional view of one of the fluid end sections making up the fluid end assembly shown in FIG. **6**, taken along line A-A.
[0018] FIG. **10** is a front perspective view of one of the housings used with one of the fluid end sections shown in FIG. **7**.
[0019] FIG. **11** is a rear perspective view of the housing shown in FIG. **10**.
[0020] FIG. **12** is a front elevational view of the housing shown in FIG. **10**.
[0021] FIG. **13** is a rear elevational view of the housing shown in FIG. **10**.
[0022] FIG. **14** is a cross-sectional view of the housing shown in FIG. **13**, taken along line B-B.
[0023] FIG. **15** is a cross-sectional view of the housing shown in FIG. **13**, taken along line C-C.
[0024] FIG. **16** is a cross-sectional view of the housing shown in FIG. **13**, taken along line D-D.
[0025] FIG. **17** is a front perspective view of the first section of the housing shown in FIG. **10**.
[0026] FIG. **18** is a rear perspective view of the first section shown in FIG. **17**.
[0027] FIG. **19** is a front perspective view of the first section shown in FIG. **17**, but the first section has a plurality of stay rods attached thereto.
[0028] FIG. **20** is a rear perspective view of the first section and stay rods shown in FIG. **19**.
[0029] FIG. **21** is a cross-sectional view of the first section and stay rods shown in FIG. **19**, taken along line E-E.
[0030] FIG. **22** is an enlarged view of area F shown in FIG. **21**.
[0031] FIG. **23** is a front perspective view of a second section of the housing shown in FIG. **10**.
[0032] FIG. **24** is a rear perspective view of the second section shown in FIG. **23**.
[0033] FIG. **25** is a front perspective view of the third section of the housing shown in FIG. **10**.
[0034] FIG. **26** is a rear perspective view of the third section shown in FIG. **25**.
[0035] FIG. **27** is a front perspective and exploded view of the housing shown in FIG. **10**.
[0036] FIG. **28** is a rear perspective and exploded view of the housing shown in FIG. **10**.
[0037] FIG. **29** is a cross-sectional view of the fluid end assembly shown in FIG. **6**, taken along line G-G.
[0038] FIG. **30** is a rear perspective view of the retention plate shown in FIGS. **9** and **29**.
[0039] FIG. **31** is a front perspective view of the retention plate shown in FIG. **30**.
[0040] FIG. **32** is a rear perspective view of the retention plate shown in FIG. **30** attached to the housing shown in FIG. **10**.
[0041] FIG. **33** is a front perspective view of the stuffing box shown in FIGS. **9** and **29**.
[0042] FIG. **34** is a rear perspective view of the stuffing box shown in FIG. **33**.
[0043] FIG. **35** is a front perspective view of the rear retainer shown in FIGS. **9** and **29**.
[0044] FIG. **36** is a rear perspective view of the rear retainer shown in FIG. **35**.
[0045] FIG. **37** is a front perspective view of the packing nut shown in FIGS. **9** and **29**.
[0046] FIG. **38** is a rear perspective view of the packing nut shown in FIG. **37**.
[0047] FIG. **39** is an enlarged cross-sectional view of the rear of a fluid end section shown in FIG. **6**, taken along line H-H.
[0048] FIG. **40** is the cross-sectional view of the fluid end section shown in FIG. **9**. The plunger is retracted, and the discharge valve is shown in a closed position.
[0049] FIG. **41** is the cross-sectional view of the fluid end section shown in FIG. **40**, but the plunger is extended within the fluid end section and the suction valve is in a closed position.
[0050] FIG. **42** is an enlarged view of area I shown in FIG. **41**.
[0051] FIG. **43** is a front perspective view of the fluid routing plug shown in FIGS. **40** and **41**.
[0052] FIG. **44** is a top plan view of the fluid routing plug shown in FIG. **43**.

[0053] FIG. **45** is a front elevational view of the fluid routing plug shown in FIG. **43**.
[0054] FIG. **46** is a rear perspective view of the fluid routing plug shown in FIG. **43**.
[0055] FIG. **47** is a rear elevational view of the fluid routing plug shown in FIG. **43**.
[0056] FIG. **48** is a cross-sectional view of the fluid routing plug shown in FIG. **47**, taken along line J-J.
[0057] FIG. **49** is a top perspective view of the fluid routing plug shown in FIG. **43**.
[0058] FIG. **50** is an enlarged view of area K shown in FIG. **49**.
[0059] FIG. **51** is a perspective cross-sectional view of the fluid routing plug shown in FIG. **44**, taken along line L-L.
[0060] FIG. **52** is a perspective cross-sectional view of the fluid routing plug shown in FIG. **44**, taken along line M-M.
[0061] FIG. **53** is a perspective cross-sectional view of the fluid routing plug shown in FIG. **44**, taken along line N-N.
[0062] FIG. **54** is a rear elevational view of the fluid routing plug shown in FIG. **43**, but one of the suction passages and one of the discharge passages are shown in phantom.
[0063] FIG. **55** is a cross-sectional view of the fluid routing plug shown in FIG. **44**, taken along line M-M.
[0064] FIG. **56** is a front elevational and conical-sectional view of the fluid routing plug shown in FIG. **43**. The conical-section is taken from line O in FIGS. **43** and **45** to line P in FIGS. **46** and **47**.
[0065] FIG. **57** is a front perspective and conical-sectional view of the fluid routing plug shown in FIG. **56**.
[0066] FIG. **58** is a rear perspective and conical-sectional view of the fluid routing plug shown in FIG. **56**.
[0067] FIG. **59** is a side elevational and conical-sectional view of the fluid routing plug shown in FIG. **56**.
[0068] FIG. **60** is a side elevational view of the hardened insert shown in FIGS. **40** and **41**.
[0069] FIG. **61** is a front perspective view of the hardened insert shown in FIG. **60**.
[0070] FIG. **62** is a front elevational view of the hardened insert shown in FIG. **60**.
[0071] FIG. **63** is a cross-sectional view of the hardened insert shown in FIG. **62**, taken along line Q-Q.
[0072] FIG. **64** is a front perspective view of the discharge valve shown in FIGS. **40** and **41**.
[0073] FIG. **65** is a rear perspective view of the discharge valve shown in FIG. **64**.
[0074] FIG. **66** is a side elevational view of the discharge valve shown in FIG. **64**.
[0075] FIG. **67** is a cross-sectional view of the discharge valve shown in FIG. **66**, taken along line R-R.
[0076] FIG. **68** is a front perspective view of the suction valve guide shown in FIGS. **40** and **41**.
[0077] FIG. **69** is a rear perspective view of the suction valve guide shown in FIG. **68**.
[0078] FIG. **70** is a front elevational view of the suction valve guide shown in FIG. **68**.
[0079] FIG. **71** is a cross-sectional view of the suction valve guide shown in FIG. **70**, taken along line S-S.
[0080] FIG. **72** is a front perspective view of the discharge plug shown in FIGS. **40** and **41**.
[0081] FIG. **73** is a rear perspective view of the discharge plug shown in FIG. **72**.
[0082] FIG. **74** is a front elevational view of the discharge plug shown in FIG. **72**.
[0083] FIG. **75** is a cross-sectional view of the discharge plug shown in FIG. **74**, taken along line T-T.
[0084] FIG. **76** is a front perspective view of the front retainer shown in FIGS. **40** and **41**.
[0085] FIG. **77** is a rear perspective view of the front retainer shown in FIG. **76**.
[0086] FIG. **78** is an enlarged view of area U shown in FIG. **41**.
[0087] FIG. **79** is a front perspective and exploded view of the fluid end section shown in FIGS. **9**, **29**, **40**, and **41**.

[0088] FIG. **80** is a rear perspective and exploded view of the fluid end section shown in FIGS. **9**, **29**, **40**, and **41**.

Alternative Embodiments

[0089] FIG. **81** is a cross-sectional view of an alternative embodiment of a fluid end section having an alternative embodiment of a fluid routing plug installed therein.

[0090] FIG. **82** is an enlarged view of area V shown in FIG. **81**.

[0091] FIG. **83** is an enlarged view of area W shown in FIG. **81**.

[0092] FIG. **84** is a top plan view of the fluid routing plug shown in FIG. **81**.

[0093] FIG. **85** is a front perspective view of the fluid routing plug shown in FIG. **84**.

[0094] FIG. **86** is a front elevational view of the fluid routing plug shown in FIG. **84**.

[0095] FIG. **87** is a cross-sectional view of the fluid routing plug shown in FIG. **86**, taken along line X-X.

[0096] FIG. **88** is a top perspective view of the fluid routing plug shown in FIG. **84**.

[0097] FIG. **89** is an enlarged view of area Y shown in FIG. **88**.

[0098] FIG. **90** is a rear elevational view of the fluid routing plug shown in FIG. **84**, but one of the suction passages and one of the discharge passages are shown in phantom.

[0099] FIG. **91** is a cross-sectional view of the fluid routing plug shown in FIG. **84**, taken along line Z-Z.

[0100] FIG. **92** is a rear perspective view of the fluid routing plug shown in FIG. **84**.

[0101] FIG. **93** is a rear elevational view of the fluid routing plug shown in FIG. **84**.

[0102] FIG. **94** is a perspective cross-sectional view of the fluid routing plug shown in FIG. **84**, taken along line AA-AA.

[0103] FIG. **95** is a perspective cross-sectional view of the fluid routing plug shown in FIG. **84**, taken along line AB-AB.

[0104] FIG. **96** is a perspective cross-sectional view of the fluid routing plug shown in FIG. **84**, taken along line AC-AC.

[0105] FIG. **97** is a front elevational and conical-sectional view of the fluid routing plug shown in FIG. **84**. The conical-section is taken from line AD in FIGS. **85** and **86** to line AE in FIGS. **92** and **93**.

[0106] FIG. **98** is a front perspective and conical-sectional view of the fluid routing plug shown in FIG. **97**.

[0107] FIG. **99** is a rear perspective and conical-sectional view of the fluid routing plug shown in FIG. **97**.

[0108] FIG. **100** is a side elevational and conical-sectional view of the fluid routing plug shown in FIG. **97**.

[0109] FIG. **101** is a cross-sectional view of another embodiment of a fluid end section having the fluid routing plug shown in FIG. **84** installed therein.

[0110] FIG. **102** is the cross-sectional view of the fluid end section shown in FIG. **101** having another embodiment of a fluid routing plug installed therein.

[0111] FIG. **103** is an enlarged view of area AF shown in FIG. **102**.

[0112] FIG. **104** is the cross-sectional view of the fluid end section shown in FIG. **81** having another embodiment of a fluid routing plug installed therein.

[0113] FIG. **105** is an enlarged view of area AG shown in FIG. **104**.

[0114] FIG. **106** is a cross-sectional view of another embodiment of a fluid end section having another embodiment of a fluid routing plug installed therein.

[0115] FIG. **107** is a top plan view of the fluid routing plug shown in FIG. **106**.

[0116] FIG. **108** is a front perspective view of the fluid routing plug shown in FIG. **107**.

[0117] FIG. **109** is a front elevational view of the fluid routing plug shown in FIG. **107**.

[0118] FIG. **110** is a cross-sectional view of the fluid routing plug shown in FIG. **109**, taken along line AH-AH.

[0119] FIG. **111** is an enlarged view of area AI shown in FIG. **106**.
[0120] FIG. **112** is an enlarged view of area AJ shown in FIG. **110**.
[0121] FIG. **113** is a cross-sectional view of another embodiment of a fluid end section having another embodiment of a fluid routing plug installed therein.
[0122] FIG. **114** is a front perspective view of the fluid routing plug shown in FIG. **113**.
[0123] FIG. **115** is a rear perspective view of the fluid routing plug shown in FIG. **114**.
[0124] FIG. **116** is a front elevational view of the fluid routing plug shown in FIG. **114**.
[0125] FIG. **117** is a cross-sectional view of the fluid routing plug shown in FIG. **116**, taken along line AK-AK.
[0126] FIG. **118** is an enlarged view of area AL shown in FIG. **117**.
[0127] FIG. **119** is an enlarged view of area AM shown in FIG. **113**.
[0128] FIG. **120** is a cross-sectional view of the fluid end section shown in FIG. **6** taken along line A-A. The fluid end section has another embodiment of a fluid routing plug installed, the plunger is fully extended and both valves are shown in the closed position.
[0129] FIG. **121** is an enlarged view of area AN shown in FIG. **120**.
[0130] FIG. **122** is a front perspective view of the fluid routing plug shown in FIG. **120**.
[0131] FIG. **123** is a front elevation view of the fluid routing plug shown in FIG. **122**.
[0132] FIG. **124** a rear perspective view of the fluid routing plug shown in FIG. **122**.
[0133] FIG. **125** is rear elevation view of the fluid routing plug shown in FIG. **122**.
[0134] FIG. **126** is a cross-sectional view of the fluid routing plug shown in FIG. **125** taken along line AO-AO.
[0135] FIG. **127** is an enlarged view of area AP shown in FIG. **126**.
[0136] FIG. **128** is an enlarged view of area AQ shown in FIG. **126**.
[0137] FIG. **129** is a top plan view of the fluid routing plug shown in FIG. **122**.
[0138] FIG. **130** is an enlarged view of area AR shown in FIG. **129**.
[0139] FIG. **131** is an enlarged view of area AS shown in FIG. **129**
[0140] FIG. **132** is an enlarged view of area AT shown in FIG. **129**.
[0141] FIG. **133** is a cross-sectional view of a suction fluid passage taken perpendicular to the longitudinal axis of the suction fluid passage.
[0142] FIG. **134** is a top, front perspective view of the fluid routing plug shown in FIG. **122**.
[0143] FIG. **135** is an enlarged view of area AU shown in FIG. **134**.
[0144] FIG. **136** is a bottom, rear perspective view of the fluid routing plug shown in FIG. **122**.
[0145] FIG. **137** is an enlarged view of area AV shown in FIG. **136**.
[0146] FIG. **138** is a cross-sectional view of another embodiment of a fluid end section having another embodiment of a fluid routing plug installed therein.
[0147] FIG. **139** is an enlarged view of area AW shown in FIG. **121**, illustrating a cross-sectional view of an insert installed in a dynamic body with deformed front and rear surfaces.
[0148] FIG. **140** is an enlarged view of area AX shown in FIG. **63**, in which insert **210** has been replaced with insert **12210** to show the pre-formed front and rear surfaces.
[0149] FIG. **141** is an enlarged view of area AW shown in FIG. **121**, in which insert **210** has been replaced by insert **12210**, showing the perpendicular surfaces of insert **12210** after installation.
[0150] FIG. **142** is a top plan view of another embodiment of a fluid routing plug.
[0151] FIG. **143** is a cross-sectional view of the fluid routing plug shown in FIG. **142**, taken along line AY-AY.
[0152] FIG. **144** is an enlarged view of area AZ shown in FIG. **143**.
[0153] FIG. **145** is a top plan view of another embodiment of a dynamic body.
[0154] FIG. **146** is a cross-sectional view of the dynamic body shown in FIG. **145**, taken along line BA-BA.
[0155] FIG. **147** is an enlarged view of area BB shown in FIG. **146**.
[0156] FIG. **148** is an enlarged view of area AW shown in FIG. **121**, in which fluid routing plug

1100 has been replaced by fluid routing plug **1300** and dynamic body **76** has been replaced by dynamic body **1376**.
[0157] FIG. **149** is a perspective view of another embodiment of a fluid routing plug.
[0158] FIG. **150** is a top plan view of the fluid routing plug shown in FIG. **149**.
[0159] FIG. **151** is a perspective view of another embodiment of a fluid routing plug.
[0160] FIG. **152** is a top plan view of the fluid routing plug shown in FIG. **151**.
[0161] FIG. **153** is a perspective view of another embodiment of a fluid routing plug.
[0162] FIG. **154** is a rear perspective view of the fluid routing plug shown in FIG. **153**.
[0163] FIG. **155** is a top plan view of the fluid routing plug shown in FIG. **153**.
[0164] FIG. **156** is a cross-sectional view of the fluid routing plug shown in FIG. **155**, taken along line BC-BC.
[0165] FIG. **157** is a cross-sectional view of the fluid routing plug shown in FIG. **156**, taken along line BD-BD. FIG. **156** is used to illustrate the location of the section line but the figure shows the entire cross-section. Only the cross-section of the suction fluid passage is shown for clarity.
[0166] FIG. **158** is a cross-sectional view of the fluid routing plug shown in FIG. **156**, taken along line BE-BE. FIG. **156** is used to illustrate the location of the section line but the figure shows the entire cross-section. Only the cross-section of the suction fluid passage is shown for clarity.
[0167] FIG. **159** is a cross-sectional view of the fluid routing plug shown in FIG. **156**, taken along line BF-BF. FIG. **156** is used to illustrate the location of the section line but the figure shows the entire cross-section. Only the cross-section of the suction fluid passage is shown for clarity.
[0168] FIG. **160** is a cross-sectional view of the fluid routing plug shown in FIG. **156**, taken along line BG-BG. FIG. **156** is used to illustrate the location of the section line but the figure shows the entire cross-section. Only the cross-section of the suction fluid passage is shown for clarity.
[0169] FIG. **161** is a cross-sectional view of the fluid routing plug shown in FIG. **156**, taken along line BH-BH. FIG. **156** is used to illustrate the location of the section line but the figure shows the entire cross-section. Only the cross-section of the suction fluid passage is shown for clarity.
[0170] FIG. **162** is a perspective view of another embodiment of a fluid routing plug.
[0171] FIG. **163** is a rear perspective view of the fluid routing plug shown in FIG. **162**.
[0172] FIG. **164** is a top plan view of the fluid routing plug shown in FIG. **162**.
[0173] FIG. **165** is a cross-sectional view of the fluid routing plug shown in FIG. **164**, taken along line BI-BI.
[0174] FIG. **166** is perspective view of another embodiment of a fluid routing plug.
[0175] FIG. **167** is a rear perspective view of the fluid routing plug shown in FIG. **166**.
[0176] FIG. **168** is a top plan view of the fluid routing plug shown in FIG. **166**.
[0177] FIG. **169** is a cross-sectional view of the fluid routing plug shown in FIG. **168**, taken along line BJ-BJ.
[0178] FIG. **170** is an enlarged view of area BK shown in FIG. **169**. The height is magnified to illustrate the described features.
[0179] FIG. **171** is an enlarged view of area AN shown in FIG. **120**, in which fluid routing plug **1000** has been replaced by fluid routing plug **1800** and insert **210** has been replaced by insert **18210**.
[0180] FIG. **172** is an enlarged view of area BL shown in FIG. **171**. Some features have been magnified to illustrate the described features.
[0181] FIG. **173** is an enlarged view of area BK shown in FIG. **169**. Fluid routing plug **1800** has been replace by fluid routing plug **1900**. The height is magnified to illustrate the described features.
[0182] FIG. **174** is an enlarged view of area BL shown in FIG. **171**. Fluid routing plug **1800** has been replaced by fluid routing plug **1900**. Insert **18210** has been replaced by insert **1910**. Some features have been magnified to illustrate the described features.
[0183] FIG. **175** is an enlarged view of area BK shown in FIG. **169**. Fluid routing plug **1800** has been replaced by fluid routing plug **2000**. The height is magnified to illustrate the described

features.
[0184] FIG. **176** is an enlarged view of area BL shown in FIG. **171**. Fluid routing plug **1800** has been replaced by fluid routing plug **2000**. Insert **18210** has been replaced by insert **20210**. Some features have been greatly magnified to illustrate the described features.
DETAILED DESCRIPTION
Fluid End Assembly
[0185] Turning now to FIGS. **3-80**, a high pressure pump **50** is shown in FIG. **3**. The pump **50** comprises a fluid end assembly **52** joined to a power end assembly **54**. The power end assembly **54** is described in more detail in U.S. patent application Ser. No. 17/884,691, authored by Keith et al., and filed on Aug. 10, 2022, the entire contents of which are incorporated herein by reference (hereinafter "the '691 application"). In alternative embodiments, the fluid end assembly **52** may be attached to other power end designs known in the art. The fluid end assembly **52** is described herein. Fluid is routed throughout the fluid end assembly using a fluid routing plug **132**. The fluid routing plug **132** and various embodiments thereof are described herein.
Fluid End Section
[0186] Turning to FIGS. **4-8**, the fluid end assembly **52** comprises a plurality of fluid end sections **56** positioned in a side-by-side relationship, as shown in FIGS. **6-8**. Each fluid end section **56** is attached to the power end assembly **54** using a plurality of stay rods **58**, as shown in FIGS. **4** and **5**. Preferably, the fluid end assembly **52** comprises five fluid end sections **56** positioned adjacent one another. In alternative embodiments, the fluid end assembly **52** may comprise more or less than five fluid end sections **56**. In operation, a single fluid end section **56** may be removed and replaced without removing the other fluid end sections **56** from the fluid end assembly **52**.
Housing of Fluid End Section
[0187] Turning to FIGS. **10-16**, each fluid end section **56** comprises a horizontally positioned housing **60** having a longitudinal axis **62** extending therethrough, as shown in FIGS. **10** and **11**. The housing **60** has opposed front and rear surfaces **64** and **66** joined by an outer intermediate surface **68**. A horizontal bore **70** is formed within the housing **60** and interconnects the front and rear surfaces **64** and **66**, as shown in FIGS. **14** and **15**. The horizontal bore **70** is sized to receive various components configured to route fluid throughout the housing **60**, as shown in FIG. **9**. The various components will be described in more detail later herein.
[0188] Continuing with FIGS. **10-16**, the housing **60** is of multi-piece construction. The housing **60** comprises a first section **72** joined to a second section **74** and a third section **76** by a plurality of first fasteners **78**, as shown in FIGS. **15** and **16**. By making the housing **60** out of multiple pieces rather than a single, integral piece, any one of the sections **72**, **74**, and **76** may be removed and replaced with a new section **72**, **74**, and **76**, without replacing the other sections. For example, if a portion of the second section **74** begins to erode or crack, the second section **74** can be replaced without having to replace the first or third sections **72** and **76**. In contrast, if the housing **60** were one single piece, the entire housing would need to be replaced, resulting in much more costly repair to the fluid end assembly **52**.
First Section of Housing
[0189] Turning to FIGS. **17-21**, the first section **72** is positioned at the front end of the housing **60** and includes the front surface **64**. During operation, fluid within the first section **72** remains at relatively the same high pressure. Thus, the first section **72** is considered the static or constant high pressure section of the housing **60**. The first section **72** is configured to be attached to a plurality of the stay rods **58**, as shown in FIGS. **19-21**. Thus, each fluid end section **56** is attached to the power end assembly **54** via the first section **72** of the housing **60**.
[0190] Continuing with FIGS. **17** and **18**, each first section **72** comprises the front surface **64** joined to a rear surface **80**. The surfaces **64** and **80** are interconnected by a portion of the outer intermediate surface **68** and a portion of the horizontal bore **70**. The outer intermediate surface **68** of the first section has the shape of a rectangular prism with a plurality of notches **82** formed within

the front surface **64**. A notch **82** is formed within each corner of the first section **72** such that the front surface **64** has a cross-sectional shape of a cross sign having radiused corners. The notches **82** are configured to receive a first end **84** of each stay rod **58**, as shown in FIG. **21**.

[0191] With reference to FIGS. **17-21**, a plurality of passages **86** are formed in the first section **72**. Each passage **86** interconnects the rear surface **80** and a medial surface **88** of the first section **72**. The medial surface **88** is defined by the plurality of notches **82**. Each passage **86** comprises a counterbore **87** that opens on the rear surface **80**, as shown in FIG. **16**, and is configured to receive a corresponding one of the stay rods **58**. When installed within the first section **72**, the first end **84** of each stay rod **58** projects from the medial surface **88** and into the corresponding notch **82**, as shown in FIG. **21**.

[0192] Continuing with FIGS. **19-21**, a threaded nut **90** is installed on the first end **84** of each stay rod **58** within each notch **82**. The nut **90** is a three-piece nut, also known as a torque nut, that facilitates the application of high torque required to properly fasten the fluid end section **56** to the power end assembly **54**. The nut **90** is described in more detail in the '691 application. In alternative embodiments, a traditional 12-point flange nut similar to the flange nut **230**, shown in FIGS. **27** and **28**, may be installed on the first end **84** of each stay rod **58** instead of the nut **90**.

[0193] Continuing with FIGS. **19-22**, a sleeve **94** is disposed around a portion of each stay rod **58** and extends between the rear surface **80** of the first section **72** and the power end assembly **54**, as shown in FIG. **3**. A dowel sleeve **93** is inserted into each counterbore **87** formed in each passage **86**, as shown in FIG. **22**. When installed therein, a portion of the dowel sleeve **93** projects from the rear surface **80** of the first section **72**. A counterbore **95** is formed within the hollow interior of the sleeve **94** for receiving the projecting end of the dowel sleeve **93**, as shown in FIG. **22**. The dowel sleeve **93** aligns the sleeve **94** and the passage **86** concentrically. Such alignment maintains a planar engagement between the rear surface **80** of the first section **72** and the sleeve **94**. When the nut **90** is torqued against the medial surface **88** of the first section **72**, the sleeve **94** abuts the rear surface **80** of the first section **72**, rigidly securing the first section **72** to the stay rod **58**.

[0194] Turning back to FIGS. **14-16**, and **18**, a plurality of threaded openings **96** are formed in the rear surface **80** of the first section **72**. The openings **96** surround an opening of the horizontal bore **70**, as shown in FIG. **18**. Each opening **96** is configured to receive a corresponding one of the first fasteners **78** used to secure the sections **72**, **74**, and **76** together, as shown in FIGS. **15** and **16**. A plurality of dowel openings **98** are also formed in the rear surface **80** adjacent the openings **96**, as shown in FIGS. **14** and **18**. The dowel openings **98** are configured to receive first alignment dowels **100**, as shown in FIG. **14**. The first alignment dowels **100** assist in properly aligning the first section **72** and the second section **74** during assembly of the housing **60**.

[0195] Continuing with FIG. **14**, a pair of upper and lower discharge bores **102** and **104** are formed within the first section **72** and interconnect the intermediate surface **68** and the horizontal bore **70**. The upper and lower discharge bores **102** and **104** shown in FIG. **14** are collinear. In alternative embodiments, the bores **102** and **104** may be offset from one another and not collinear. Each bore **102** and **104** may include a counterbore **106** that opens on the intermediate surface **68**. Each counterbore **106** is sized to receive a portion of a discharge fitting adapter **504**, as shown in FIG. **9**. The fitting adapter **504** spans between the discharge bore **102** or **104** and a discharge fitting **108** attached to the outer intermediate surface **68** of the first section **72**.

[0196] Continuing with FIG. **14**, a groove **110** may be formed in the side walls of the counterbore **106** for receiving a seal **112**. The seal **112** engages an outer surface of the fitting adapter **504** to prevent fluid from leaking between the first section **72** and the discharge fitting **108**, as shown in FIG. **9**.

[0197] With reference to FIGS. **17** and **18**, a plurality of threaded openings **114** are formed in the intermediate surface **68** and surrounding the opening of the upper and lower discharge bores **102** and **104**. The threaded openings **114** are configured to receive a plurality of threaded fasteners **116** configured to secure a discharge fitting **108** to the first section **72**, as shown in FIG. **9**.

[0198] Continuing with FIGS. **14** and **15**, the walls surrounding the horizontal bore **70** within the first section **72** and positioned between the front surface **64** and the upper and lower discharge bores **102** and **104** are sized to receive a front retainer **118** and a discharge plug **120**, as shown in FIG. **9**. The discharge plug **120** seals fluid from leaking from the front surface **64** of the housing **60**, and the front retainer **118** secures the discharge plug **120** within the first section **72** of the housing **60**.

[0199] Continuing with FIGS. **14** and **15**, internal threads **122** are formed in the walls of the first section **72** for mating with external threads **124**, shown in FIGS. **76** and **77**, formed on an outer surface of the front retainer **118**. In contrast, an outer surface of the discharge plug **120** faces flat walls of the first section **72**. A small amount of clearance may exist between the plug **120** and the walls of the first section **72**.

[0200] Continuing with FIGS. **14** and **15**, a groove **125** may be formed in such walls for receiving a seal **126** configured to engage an outer surface of the discharge plug **120**, as shown in FIG. **9**. The seal **126** prevents fluid from leaking around the discharge plug **120** during operation. A locating cutout **128** may further be formed in the walls that is configured to receive a locating dowel pin **130**. As will be described later herein, the locating dowel pin **130** is used to properly align the discharge plug **120** within the housing **60**.

[0201] Continuing with FIGS. **14** and **15**, the walls surrounding the horizontal bore **70** and positioned between the upper and lower discharge bores **102** and **104** and the rear surface **80** of the first section **72** are sized to receive a portion of a fluid routing plug **132**, as shown in FIG. **9**. This area of the walls surrounding the horizontal bore **70** includes a counterbore **134** that opens on the rear surface **80**. The counterbore **134** is sized to receive a wear ring **136**, as shown in FIG. **9**. The wear ring **136** has an annular shape and is configured to engage a first seal **386** installed within an outer surface of the fluid routing plug **132**, as shown in FIG. **9**. In alternative embodiments, the first section **72** may not include the counterbore **134** or the wear ring **136** and instead may be sized to directly engage the first seal **386** installed within the fluid routing plug **132**.

[0202] Continuing with FIG. **9**, in addition to the above mentioned components, the first section **72** is also configured to house a discharge valve **138**. The components discussed above and installed within the first section **72** will be described in more detail later herein.

Second Section of Housing

[0203] Turning to FIGS. **23** and **24**, the second section **74** of the housing **60** is configured to be positioned between the first and third sections **72** and **76** and has a cylindrical cross-sectional shape. During operation, fluid pressure within the second section **74** remains at relatively the same pressure. The pressure is lower than that within the first section **72**. Thus, the second section **74** may be referred to as the static or constant low pressure section of the housing **60**. The second section **74** comprises opposed front and rear surfaces **140** and **142** joined by a portion of the outer intermediate surface **68** and a portion of the horizontal bore **70**.

[0204] Continuing with FIGS. **15**, **16**, **23**, and **24**, a plurality of passages **144** are formed in the second section **74**. The passages **144** surround the horizontal bore **70** and interconnect the front and rear surfaces **140** and **142**, as shown in FIGS. **15** and **16**. Each passage **144** is configured to receive a corresponding one of the first fasteners **78** used to secure the sections **72**, **74**, and **76** of the housing **60** together.

[0205] Continuing with FIGS. **14**, **23**, and **24**, a plurality of dowel openings **146** are formed in the front surface **140** of the second section **74**, as shown in FIG. **23**. The dowel openings **146** align with the dowel openings **98** formed in the rear surface **80** of the first section **72** and are configured to receive a portion of the first alignment dowels **100**, as shown in FIG. **14**. Likewise, a plurality of dowel openings **148** are formed in the rear surface **142** of the second section **74**, as shown in FIG. **24**. The dowel openings **148** are configured to receive a portion of second alignment dowels **150**, as shown in FIG. **14**. The second alignment dowels **150** are configured to align the second section **74** and the third section **76** during assembly.

[0206] Continuing with FIGS. **14**, **15**, **23** and **24**, a first annular groove **152** is formed in the front surface **140** of the second section **74** such that it surrounds an opening of the horizontal bore **70**, as shown in FIG. **23**. The first groove **152** is positioned between the horizontal bore **70** and the plurality of passages **144** and is configured to receive a first seal **154**, as shown in FIGS. **14** and **15**. Likewise, a second annular groove **156** is formed in the rear surface **142** of the second section **74** and positioned between the horizontal bore **70** and the plurality of passages **144**, as shown in FIG. **24**. The second groove **156** is configured to receive a second seal **158**, as shown in FIGS. **14** and **15**. The seals **154** and **158** shown in FIGS. **14** and **15** are O-rings. The seals **154** and **158** prevent fluid from leaking between the first and second sections **72** and **74** and between the second and third sections **74** and **76** during operation.

[0207] Continuing with FIGS. **14**, **23**, and **24**, a pair of upper and lower suction bores **160** and **162** are formed within the second section **74** and interconnect the intermediate surface **68** and the horizontal bore **70**. The upper and lower suction bores **160** and **162** shown in FIG. **14** are collinear. In alternative embodiments, the bores **160** and **162** may be offset from one another and not collinear.

[0208] Continuing with FIGS. **9** and **14**, the suction bores **160** and **162** are each configured to receive a suction conduit **166**, as shown in FIG. **9**. The suction conduit **166** comprises a first connection member **164** configured to mate with the housing **60**. Each suction bore **160** and **162** opens into a counterbore **168** sized to receive a portion of the first connection member **164**. Internal threads **170** are formed in a portion of the walls surrounding the counterbore **168** for mating with external threads **172**, shown in FIG. **78**, formed on the first connection member **164**.

[0209] Continuing with FIGS. **9** and **14**, a groove **174** is formed in the walls surrounding the counterbore **168** and configured to receive a seal **176**, as shown in FIG. **9**. The seal **176** engages an outer surface of the first connection member **164** to prevent fluid from leaking from the housing **60** during operation. The suction conduits **166** will be described in more detail later herein.

[0210] Continuing with FIGS. **9**, **14**, and **15**, the walls surrounding the horizontal bore **70** within the second section **74** are configured to receive a majority of the fluid routing plug **132**, as shown in FIG. **9**. A small amount of clearance may exist between the walls of the second section **74** and an outer surface of the fluid routing plug **132**.

Third Section of Housing

[0211] Turning to FIGS. **14-16**, **25** and **26**, the third section **76** of the housing **60** is positioned at the rear end of the housing **60** and includes the rear surface **66**. The third section **76** has a generally cylindrical cross-sectional shape. Fluid pressure within the third section **76** varies during operation. Thus, the third section **76** may be referred to as the dynamic or variable pressure section of the housing **60**.

[0212] Continuing with FIGS. **25** and **26**, the third section **76** comprises a front surface **178** joined to the rear surface **66** of the housing **60** by a portion of the outer intermediate surface **68** and a portion of the horizontal bore **70**. The outer intermediate surface **68** of the third section **76** varies in diameter such that the third section **76** comprises a front portion **180** joined to a rear portion **182**.

[0213] Continuing with FIGS. **15**, **16**, **25**, and **26**, the front portion **180** has a constant outer diameter and has a plurality of passages **184** formed therein. The passages **184** interconnect the front surface **178** and a medial surface **186** of the third section **76**. The passages **184** align with the plurality of passages **144** formed in the second section **74** and the threaded openings **96** formed in the first section **72** of the housing **60**, as shown in FIGS. **14** and **15**. The passages **184** are configured to receive the first fasteners **78** used to secure the sections **72**, **74**, and **76** together.

[0214] Continuing with FIGS. **14** and **25**, a plurality of dowel openings **188** are formed in the front surface **178** of the third section **76**, as shown in FIG. **25**. The dowel openings **188** are configured to receive a portion of second alignment dowels **150**, as shown in FIG. **14**. The second alignment dowels **150** are configured to properly align the third section **76** with the second section **74** during assembly.

[0215] Continuing with FIGS. **25** and **26**, the rear portion **182** of the third section **76** comprises a neck **190** joined to a shoulder **192**. The neck **190** interconnects the front portion **180** and the shoulder **192**. The shoulder **192** includes the rear surface **66** of the housing **60**. The neck **190** has a smaller outer diameter than that of the front portion **180** and the shoulder **192** to provide clearance for the plurality of passages **184** formed in the front portion **180**.

[0216] With reference to FIGS. **25** and **26**, a plurality of first threaded openings **194** are formed in the rear surface **66** of the third section **76**. The first threaded openings **194** are configured to receive a plurality of second fasteners **196**, as shown in FIG. **29**. The second fasteners **196** are configured to secure a stuffing box **198** and a rear retainer **200** to the third section **76** of the housing **60**, as shown in FIG. **29**. The stuffing box **198** and the rear retainer **200** will be described in more detail later herein.

[0217] With reference to FIGS. **26** and **32**, a plurality of second threaded openings **202** are also formed in the rear surface **66** of the third section **76**, as shown in FIG. **26**. The second threaded openings **202** are configured to receive a plurality of third fasteners **204**. The third fasteners **204** are configured to secure a retention plate **206** to the rear surface of the housing **60**, as shown in FIG. **32**. The retention plate **206** will be described in more detail later herein. A plurality of dowel openings **207** are also formed in the rear surface **66** of the third section **76**. The dowel openings **207** are configured to receive third alignment dowels **242**, as shown in FIG. **39**.

[0218] Turning back to FIGS. **9**, **14**, **15**, and **25**, a counterbore **208** is formed in the walls surrounding the horizontal bore **70** within the third section **76** and opens on the front surface **178**. The counterbore **208** is configured to receive a hardened insert **210**, as shown in FIG. **9**. The insert **210** will be described in more detail later herein. The insert **210** engages portions of the fluid routing plug **132** when the fluid routing plug **132** is installed within the housing **60**, as shown in FIG. **9**. The walls surrounding the horizontal bore **70** between the counterbore **208** and the medial surface **186** of the third section **76** are further configured to receive a suction valve guide **212**. A suction valve **214** is also installed within the third section **76** of the housing **60**. The suction valve **214** and suction valve guide **212** will be described in more detail later herein.

[0219] Continuing with FIGS. **9**, **14**, **15**, **25**, and **26**, the walls surrounding the horizontal bore **70** within the neck **190** of the rear portion **182** are sized to receive at least a portion of a reciprocating plunger **216**, as shown in FIG. **9**. The portion of the horizontal bore **70** extending through the neck **190** has a uniform diameter and opens into a first counterbore **218** formed in the shoulder **192**, as shown in FIGS. **14** and **15**. The first counterbore **218** is sized to receive a portion of the stuffing box **198**, as shown in FIG. **9**. The first counterbore **218** opens into a second counterbore **220**, of which opens on the rear surface **66** of the housing **60**, as shown in FIGS. **14**, **15**, and **26**. The second counterbore **220** is sized to receive a wear ring **222** and a seal **224**, as shown in FIG. **9**. The wear ring **222** and the seal **224** each have an annular shape. When such components are installed within the housing **60**, the wear ring **222** surrounds the seal **224**, and the seal **224** engages an outer surface of the stuffing box **198**.

Assembly of Housing

[0220] Turning to FIGS. **27-29**, the housing **60** is assembled by threading a first end **226** of each of the first fasteners **78** into a corresponding one of the threaded openings **96** formed in the first section **72**. Once installed therein, the first fasteners **78** project from the rear surface **80** of the first section **72**. The second and third sections **74** and **76** may then be slid onto the fasteners **78** projecting from the first section **72** using the corresponding passages **144** and **184**. The first and second alignment dowels **100** and **150** help to further align the sections **72**, **74**, and **76** together during assembly.

[0221] Continuing with FIGS. **27-29**, when the second and third sections **74** and **76** are installed on the fasteners **78**, a second end **228** projects from the medial surface **186** of the third section **76**, as shown in FIG. **29**. A flange nut **230** is installed on the second end **228** and torqued against the medial surface **186**, tightly securing the sections **72**, **74**, and **76** together. When the housing **60** is assembled, a footprint of the rear surface **142** of the second section **74** is entirely within a footprint of the rear surface **80** of the first section **72**, as shown in FIG. **29**.

[0222] Continuing with FIGS. **27-29**, the first fastener **78** shown in the figures is a threaded stud. In alternative embodiments, other types of fasteners known in the art may be used instead of a threaded stud. For example, screws or bolts may be used to secure the sections together. In further alternative embodiments, the nut may comprise the three-piece nut **90**, shown in FIGS. **19-21**.

[0223] Continuing with FIGS. **27-29**, to remove a section **72**, **74**, or **76**, the nut **230** is unthreaded from the second end **228** of each first fastener **78**. The sections **72**, **74**, and **76** may then be pulled apart, as needed. If the first section **72** is being replaced, the first fasteners **78** are also unthreaded from the threaded openings **96**. The components installed within the housing **60** may also be removed, as needed, prior to disassembling the housing **60**.

Components Attached to Rear Surface of Housing

[0224] Turning to FIGS. **29-32**, in addition to the housing **60**, the fluid end section **56** comprises a plurality of components attached to the rear surface **66** of the housing **60**. Such components are configured to receive the plunger **216**. The various components include the retention plate **206**, the stuffing box **198**, and the rear retainer **200**, previously mentioned. The components further comprise a plunger packing **300**, and a packing nut **290**.

Retention Plate

[0225] Continuing with FIGS. **29-32**, the retention plate **206** has a cylindrical cross-sectional shape and is sized to cover the rear surface **66** of the housing **60** and the wear ring **222** and the seal **224**, as shown in FIG. **29**. The retention plate **206** holds the wear ring **222** and the seal **224** within the housing **60** in the event the stuffing box **198** needs to be removed.

[0226] Continuing with FIGS. **29-32**, the retention plate **206** comprises opposed front and rear surface **237** and **238** joined by a central opening **239** formed therein. A plurality of first passages **234** are formed in the retention plate **206** and surround the central opening **239** of the plate **206**. The first passages **234** align with the first threaded openings **194** formed in the rear surface **66** of the housing **60** and are configured to receive the plurality of second fasteners **196**.

[0227] Continuing with FIGS. **30-32**, a plurality of second passages **236** are also formed in the retention plate **206**. The second passages **236** align with the second threaded openings **202** formed in the rear surface **66** of the housing **60** and are configured to receive the third fasteners **204**, as shown in FIG. **32**. A third fastener **204** is threaded into one of the second threaded openings **202** and turned until it sits flush with the rear surface **238** of the retention plate **206**, as shown in FIG. **32**.

[0228] Continuing with FIGS. **30-32**, a plurality of dowel openings **240** are formed in the retention plate **206** for receiving third alignment dowels **242**, as shown in FIG. **39**. The third alignment dowels **242** assist in properly aligning the retention plate **206** and the stuffing box **198** on the housing **60** during assembly.

[0229] Turning back to FIG. **29**, since fluid does not contact the retention plate **206** during operation, the retention plate **206** may be made of a different and less costly material than that of the housing **60** or the stuffing box **198**. For example, the retention plate **206** may be made of alloy steel, while the housing **60** and stuffing box **198** are made of stainless steel.

Stuffing Box

[0230] Turning to FIGS. **29**, **33**, **34**, and **39**, the stuffing box **198** comprises opposed front and rear surfaces **244** and **246** joined by an outer intermediate surface **248** and a central passage **250** formed therein. The stuffing box **198** further comprises a front portion **252** joined to a rear portion **254**. The front portion **252** has a smaller outer diameter than the rear portion **254** such that a medial surface **256** is formed between the front and rear surfaces **244** and **246**. The front portion **252** includes the front surface **244** of stuffing box **198**, and the rear portion **254** includes the rear surface **246** of the stuffing box **198**. An internal shoulder **272** is formed within the walls surrounding the central passage **250** within the rear portion **254** of the stuffing box **198**.

[0231] Continuing with FIGS. **33**, **34**, and **39**, a plurality of passages **258** are formed within the rear portion **254** of the stuffing box **198** and interconnect the medial surface **256** and the rear surface **246**. The passages **258** are configured to align with the plurality of first passages **234** formed in the retention plate **206** and the plurality of first threaded openings **194** formed in the rear surface **66** of the housing **60**, as shown in FIG. **29**.

[0232] Continuing with FIGS. **33**, **34**, and **39**, a plurality of dowel openings **260** may be formed in the medial surface **256** of the stuffing box **198**. The dowel openings **260** are configured to receive at least a portion of the third alignment dowels **242** to properly align the stuffing box **198** on the retention plate **206** and the housing **60** during assembly, as shown in FIG. **39**. Likewise, a plurality of dowel openings **268** may be formed in the rear surface **246** of the stuffing box **198** for receiving fourth alignment dowels **270**, as shown in FIG. **39**. The fourth alignment dowels **270** assist in properly aligning the rear retainer **200** on the stuffing box **198** during assembly.

[0233] Continuing with FIG. **39**, the stuffing box **198** is installed within the third section **76** of the housing **60** such that the front portion **252** is disposed within the horizontal bore **70** and the medial surface **256** abuts the rear surface **238** of the retention plate **206**. The outer intermediate surface **248** of the front portion **252** of the stuffing box **198** engages the seal **224**. The seal **224** prevents fluid from leaking between the housing **60** and the stuffing box **198**.

[0234] Continuing with FIG. **39**, during operation, the seal **224** wears against the outer intermediate surface **248** of the front portion **252**. Should the front portion **252** begin to erode, the stuffing box **198** may be removed and replaced with a new stuffing box **198**. Likewise, the seal **224** wears against the wear ring **222** during operation. The wear ring **222** is preferably made of a harder and more wear resistant material than the housing **60**, such as tungsten carbide. Should the wear ring **222** begin to erode, the wear ring **222** can be removed and replaced with a new wear ring **222**. Trapping the seal **224** between replaceable parts protects the housing **60** over time.

Rear Retainer

[0235] Turning to FIGS. **29**, **35**, **36**, and **39**, the rear retainer **200** comprises opposed front and rear surfaces **276** and **278** joined by an outer intermediate surface **280** and a central passage **282** formed therein. A plurality of passages **284** are formed in the rear retainer **200** and surround the central passage **282**. The passages **284** interconnect the front and rear surfaces **276** and **278** of the rear retainer **200** and are configured to align with the passages **258** formed in the rear portion **254** of the stuffing box **198**, as shown in FIG. **29**. A plurality of dowel openings **286** are formed in the front surface **276** of the rear retainer **200** for receiving a portion of the fourth alignment dowels **270**, as shown in FIG. **39**.

[0236] Continuing with FIGS. **35**, **36**, **39**, and **40**, an internal shoulder **279** is formed within the walls surrounding the central passage **282** of the rear retainer **200**. Internal threads **288** are formed in the walls surrounding the central passage **282** and positioned between the internal shoulder **279** and the rear surface **278**. The internal threads **288** are configured to receive a packing nut **290**, as shown in FIG. **39**. The walls positioned between the internal shoulder **279** and the front surface **276** are flat and include one or more lube ports **292**. The lube port **292** interconnects the central passage **282** and the outer intermediate surface **280** of the rear retainer **200**, as shown in FIG. **40**.

Plunger Packing and Packing Nut

[0237] Continuing with FIG. **39**, fluid is prevented from leaking around the plunger **216** during operation by a plunger packing **300**. The plunger packing **300** is installed within the stuffing box **198** and comprises a plurality of packing seals **302** sandwiched between first and second metal rings **304** and **306**. The first metal ring **304** abuts the internal shoulder **272** formed within the stuffing box **198** and the second metal ring **306** extends into the central passage **282** formed in the rear retainer **200**. The second metal ring **306** is known in the art as a “lantern ring”. One or more passages **303**, shown in FIG. **29**, may be formed in the second metal ring **306** and fluidly connect with the one or more lube ports **292** formed in the rear retainer **200**. During operation, oil used to lubricate the plunger **216** and plunger packing **300** is supplied through the lube port **292** and second metal ring **306**.

[0238] With reference to FIGS. **37-39**, the plunger packing **300** is retained within the stuffing box **198** and the rear retainer **200** using the packing nut **290**. The packing nut **290** comprises opposed front and rear surfaces **308** and **310** joined by an outer intermediate surface **312** and a central passage **314** formed therein. External threads **316** are formed in a portion of the outer intermediate surface **312** for engaging the internal threads **288** formed in the rear retainer **200**, as shown in FIG. **39**. When the packing nut **290** is installed within the rear retainer **200**, the front surface **308** of the packing nut **290** engages and compresses the plunger packing **300**, as shown in FIG. **39**. When compressed, the packing seals **302** of the plunger packing **300** tightly seal against an outer surface of the plunger **216**.

[0239] Continuing with FIG. **39**, during operation, the packing nut **290** may be tightened, as needed, to ensure adequate compression of the packing seals **302** against the plunger **216**. At least a portion of the packing nut **290** projects from the rear surface **278** of the rear retainer **200** to provide clearance to turn the packing nut **290**, as needed. The central passage **314** formed in the packing nut **290** is sized to closely receive the plunger **216**. A groove **318** may be formed in the walls surrounding the central passage **314** for receiving a seal **320**. The seal **320** shown in FIG. **39** is an O-ring. The seal **320** prevents fluid from leaking around the plunger **216** during operation.

Assembly of Components on Rear Surface of Housing

[0240] Turning back to FIG. **29**, the front surface **276** of the rear retainer **200** abuts the rear surface **246** of the stuffing box **198** such that the plurality of passages **284** align with the plurality of passages **258** formed in the stuffing box **198**. A second fastener **196** is installed within a corresponding one of the aligned first threaded openings **194** and passages **234**, **258**, and **284**. A first end **294** of the second fastener **196** threads into the first threaded opening **194** and a second end **296** projects from the rear surface **278** of the rear retainer **200**. A nut **298** is threaded onto the second end **296** and torqued against the rear surface **278**, tightly securing the stuffing box **198** and the rear retainer **200** to the third section **76** of the housing **60**.

[0241] Continuing with FIG. **29**, the nut **298** shown in the figures is 12-point flange nut. In alternative embodiments, the nut may comprise the three-piece nut **90**, shown in FIGS. **19-21**. The second fastener **196** shown in the figures is a threaded stud. In alternative embodiments, the second fastener **196** may comprise other fasteners known in the art, such as a bolt or screw.

[0242] Continuing with FIG. **29**, the stuffing box **198** and rear retainer **200** are attached to the housing **60** after the retention plate **206** has first been attached to the rear surface **66** of the housing **60**. The plunger packing **300** may be installed within stuffing box **198** either before or after the stuffing box **198** is attached to the housing **60**. After all the components are assembled, the packing nut **290** is threaded into the rear retainer **200** until it engages the plunger packing **300**.

[0243] With reference to FIGS. **29** and **39**, when the retention plate **206**, the stuffing box **198**, and rear retainer **200** are attached to the housing **60**, the central opening **239** of the retention plate **206** and the central passages **250** and **282** of the stuffing box **198** and the rear retainer **200** form an extension of the horizontal bore **70**. Likewise, the interior of the plunger packing **300** and the central passage **314** of the packing nut **290** also form extensions of the horizontal bore **70**. The plunger **216** is installed within the fluid end section **56** through the rear surface **310** of the packing nut **290**. During operation, the plunger **216** reciprocates within the horizontal bore **70**, creating the variance in fluid pressure within the fluid end section **56** during operation.

[0244] With reference to FIGS. **3**, **39**, and **40**, during operation, reciprocal movement of the plunger **216** is driven by a pony rod **322** installed within the power end assembly **54**. A clamp **324** secures the plunger **216** to the pony rod **322** such that the plunger **216** and pony rod **322** move in unison.

Components Installed within the Housing

[0245] Turning to FIGS. **40-80**, the various internal components of the housing **60** will now be described in more detail. Fluid is routed throughout the housing **60** by the fluid routing plug **132**. The timing of movement throughout the fluid routing plug **132** is controlled by the suction valve

214 and the discharge valve **138**. Movement of the valves **214** and **138** is guided by the suction valve guide **212** and the discharge plug **120**.

Fluid Routing Plug

[0246] Turning to FIGS. **40-59**, the fluid routing plug **132** comprises a body **330** having a suction surface **332** and an opposed discharge surface **334** joined by an outer intermediate surface **336**. A central longitudinal axis **338** extends through the body **330** and the suction and discharge surfaces **332** and **334**. When the fluid routing plug **132** is installed within the housing **60**, at least a portion of the discharge surface **334** is positioned within the first section **72** of the housing **60**, and at least a portion of the suction surface **332** is positioned within the third section **76** of the housing **60**, as shown in FIGS. **40** and **41**.

[0247] Continuing with FIGS. **43-59**, the body **330** further comprises a plurality of suction fluid passages **340**. The suction passages **340** interconnect the intermediate surface **336** and the suction surface **332** of the body **330**, as shown in FIG. **48**. The connection is formed within a blind bore **342** formed within the suction surface **332** of the body **330**. During operation, fluid entering the housing **60** through the suction bores **160** and **162** flows into the suction passages **340** of the fluid routing plug **132** and into the blind bore **342**. From there, fluid flows towards the suction surface **332** of the body **330** and out of the fluid routing plug **132**. Three suction fluid passages **340** are shown in FIGS. **43-59**. In alternative embodiments, more or less than three suction fluid passages **340** may be formed within the body **330**.

[0248] Continuing with FIGS. **49** and **50**, each suction fluid passage **340** has a generally oval or tear drop cross-sectional shape. An opening **344** of each suction fluid passage **340** on the intermediate surface **336** comprises a first side wall **346** joined to a second side wall **348** by first and second ends **350** and **352**. The first and second side walls **346** and **348** are straight lines of equal length S**1**, and the first and second ends **350** and **352** are circular arcs, as shown in FIG. **50**.

[0249] Continuing with FIG. **50**, the first end **350** of the opening **344** has a radius of R**1** with a center at C**1**, and the second end **352** has a radius of R**2** with a center at C**2**. The first end **350** is larger than the second end **352** such that R**1**>R**2**. The first and second side walls **346** and **348** are tangent to the first and second ends **350** and **352** and have an included angle, σ**1**.

[0250] Continuing with FIG. **50**, the opening **344** has a centerline **354** that connects the centers C**1** and C**2** of the first and second ends **350** and **352**. The centerline **354** has a length E**1** and is parallel with the central longitudinal axis **338**. A cross-sectional shape of each suction passage **340** throughout the length of the body **330** corresponds with the shape of each opening **344**, as shown in FIG. **55**. Each suction passage **340** is sized and shaped to maximize fluid flow through the passage **340** and minimize fluid turbulence and stress to the body **330** of the fluid routing plug **132**.

[0251] With reference to FIGS. **55** and **56**, each suction fluid passage **340** extends between the blind bore **342** and the intermediate surface **336** such that each suction passage **340** comprises a longitudinal axis **356**. The longitudinal axis **356** extends through the center C**1** of the first end **350** of the opening **344** and intersects the central longitudinal axis **338**, as shown in FIG. **56**.

[0252] Continuing with FIGS. **43-59**, the body **330** further comprises a plurality of discharge fluid passages **360**. The discharge fluid passages **360** interconnect the suction surface **332** and the discharge surface **334** of the body **330** and do not intersect any of the suction fluid passages **340**. Rather, the discharge and suction fluid passages **360** and **340** are in a spaced-relationship. In operation, fluid exiting the body **330** at the suction surface **332** is subsequently forced into the discharge fluid passages **360**, towards the discharge surface **334** of the body **330**, and out of the fluid routing plug **132**. Three discharge fluid passages **360** are shown in FIGS. **43-59**. In alternative embodiments, more or less than three discharge fluid passages **360** may be formed within the body **330**.

[0253] Continuing with FIGS. **43**, **46**, and **48**, the suction surface **332** of the body **330** comprises an outer rim **362** joined to the blind bore **342** by a tapered seating surface **366**, as shown in FIGS. **46** and **48**. Likewise, the discharge surface **334** comprises an outer rim **368** joined to a central base

370 by a tapered seating surface **372**, as shown in FIGS. **43** and **48**.

[0254] Continuing with FIGS. **43**, **46**, and **48**, each discharge passage **360** opens at a first opening **374** on the outer rim **362** of the suction surface **332** and opens at a second opening **376** on the central base **370** of the discharge surface **334**. The second openings **376** surround a blind bore **378** formed in the central base **370** of the discharge surface **334**. The blind bore **378** is configured to engage a tool used to grip the fluid routing plug **132**, as needed. For example, the walls of the blind bore **378** may be threaded. The central base **370** may also be slightly recessed from the tapered seating surface **372** such that a small counterbore **380** is created. The counterbore **380** helps further reduce any turbulence of fluid exiting the second openings **376**.

[0255] Continuing with FIG. **54**, a position of the first and second openings **374** and **376** of each discharge passage **360** may be determined relative to a plane containing a line **382** that is perpendicular to the central longitudinal axis **338**. The first opening **374**, when projected onto the plane, is positioned at a first distance F**1** from the central longitudinal axis **338** and at a first angle φ**1** relative to the line **382**. The second opening **376**, when projected onto the plane, is positioned at a second distance F**2** from the central longitudinal axis **338** and at a second angle φ**2** relative to the line **382**.

[0256] The first and second distances F**1** and F**2** shown in FIG. **54** are different. Likewise, the first and second angles φ**1** and φ**2** shown in FIG. **54** are different. In alternative embodiments, the first and second angles φ**1** and φ**2** may be different, but the first and second distances F**1** and F**2** may be the same. In further alternative embodiments, the first and second angles φ**1** and φ**2** may be the same, but the first and second distances F**1** and F**2** may be different. In even further alternative embodiments, the first and second distances F**1** and F**2** may be the same, and the first and second angles φ**1** and φ**2** may be the same.

[0257] With reference to FIGS. **51-53** and **56-59**, each discharge fluid passage **360** has an arced cross-sectional shape. The length of the arc may gradually increase between the suction and discharge surfaces **332** and **334**, as shown in FIGS. **51-53**. In alternative embodiments, the discharge fluid passages **360** may have different shapes and sizes. Turning back to FIGS. **44** and **48**, a first annular groove **384** is formed in the outer intermediate surface **336** of the body **330** for housing the first seal **386**. The first groove **384** is positioned adjacent the discharge surface **334** and is characterized by two sides walls **388** joined by a base **390**. When the fluid routing plug **132** is installed within the housing **60**, the first seal **386** engages an inner surface of the wear ring **136** installed within the first section **72** of the housing **60**, as shown in FIGS. **40** and **41**. During operation, the first seal **386** wears against the wear ring **136**. If the wear ring **136** begins to erode, the wear ring **136** may be removed and replaced with a new wear ring **136**. The wear ring **136** has an annular shape and may be made of a harder and more wear resistant material than the housing **60**. For example, the housing **60** may be made of stainless steel and the wear ring **136** is made of tungsten carbide.

[0258] With reference to FIGS. **42**, **44**, and **48**, a second annular groove **392** is formed in the outer intermediate surface **336** of the body **330** for housing a second seal **394**. The second groove **392** is positioned adjacent the suction surface **332** and is characterized by a plurality of side walls **396** joined by a base **398**, as shown in FIG. **42**. Four side walls **396** are shown in FIG. **42** such that the groove **392** has a rounded shape. When the fluid routing plug **132** is installed within the housing **60**, the second seal **394** engages an inner intermediate surface **420** of the hardened insert **210**, as shown in FIG. **42**. During operation, the second seal **394** wears against the insert **210**. If the insert **210** begins to erode, the insert **210** may be removed and replaced with a new insert **210**.

[0259] Continuing with FIGS. **42**, **44**, and **48**, the outer intermediate surface **336** of the body **330** further comprises an annular shoulder **400** formed in the body **330**. The shoulder **400** is positioned between the opening **344** of the suction passages **340** and the second groove **392**. When the fluid routing plug **132** is installed within the housing **60**, the shoulder **400** abuts a front surface **416** of the insert **210**, as shown in FIG. **42**. Axial movement of the fluid routing plug **132** towards the rear surface **66** of the housing **60** is prevented by the engagement between the shoulder **400** and the insert **210**. During operation, the shoulder **400** may wear against the insert **210**. If either feature begins to wear, the fluid routing plug **132** and/or the insert **210** may be removed and replaced with a new fluid routing plug **132** and/or insert **210**.

[0260] Continuing with FIGS. **40**, **41**, **44**, and **48**, the outer intermediate surface **336** of the body **330** adjacent the first groove **384** is characterized as a first cylindrical surface **404**. Likewise, the outer intermediate surface **336** adjacent the annular shoulder **400** is characterized as a second cylindrical surface **406**. The first cylindrical surface **404** has a maximum outer diameter that is equal or almost equal to a maximum outer diameter of the second cylindrical surface **406**. The surfaces **404** and **406** are configured to closely face the walls surrounding the horizontal bore **70** within the second section **74** of the housing **60**, as shown in FIGS. **40** and **41**.

[0261] Continuing with FIGS. **40**, **41**, **44**, **48**, and **49**, the outer intermediate surface **336** of the body **330** further comprises a first bevel **408** joined to a transition surface **410** formed in the body **330**. The first bevel **408** and the transition surface **410** are positioned between the first cylindrical surface **404** and the openings **344** of the suction passages **340**. The outer intermediate surface **336** of the body **330** slowly tapers outward from the transition surface **410** to the second cylindrical surface **406**.

[0262] Continuing with FIGS. **40** and **41**, when the fluid routing plug **132** is installed within the housing **60**, the first bevel **408** provides clearance between the outer intermediate surface **336** of the fluid routing plug **132** and an opening of the suction bores **160** and **162**. Such clearance gives way to an annular fluid channel **412** formed between the housing **60** and the fluid routing plug **132**. The shape of the outer intermediate surface **336** of the fluid routing plug **132** between the first and second cylindrical surfaces **404** and **406** helps direct fluid flowing from the suction bores **160** and **162** into the openings **344** of the suction passages **340** while minimizing fluid turbulence.

[0263] Turning back to FIG. **42**, the outer intermediate surface **336** of the body **330** further comprises a second bevel **414** formed in the body **330**. The second bevel **414** is positioned between the suction surface **332** and the second groove **392**. The second bevel **414** provides clearance to help install the fluid routing plug **132** within the housing **60** and the insert **210**.

Hardened Insert

[0264] With reference to FIGS. **42** and **60-63**, the insert **210** has an annular shape and comprises opposed front and rear surfaces **416** and **418** joined by inner and outer intermediate surfaces **420** and **422**. The insert **210** further comprises a first bevel **426** formed in the inner intermediate surface **420** adjacent the front surface **416**, as shown in FIG. **63**. The first bevel **426** provides clearance to assist in installing the fluid routing plug **132** within the insert **210** within the housing **60**, as shown in FIG. **42**. The insert **210** also comprises a second bevel **424** formed in the outer intermediate surface **422** adjacent the rear surface **418**. The second bevel **424** provides clearance to assist in installing the insert **210** within the counterbore **208** formed in the third section **76** of the housing **60**, as shown in FIG. **42**. The insert **210** is made of a harder and more wear resistant material than the housing **60**. For example, if the housing **60** is made of stainless steel, the insert **210** may be made of tungsten carbide.

Suction and Discharge Valves

[0265] With reference to FIGS. **40**, **41**, and **64-67**, the flow of fluid throughout the housing **60** and the fluid routing plug **132** is regulated by the suction and discharge valves **214** and **138**. The suction valve **214** is configured to engage the suction surface **332**, and the discharge valve **138** is configured to engage the discharge surface **334** of the fluid routing plug **132** such that the surfaces **332** and **334** function as valve seats. The valves **214** and **138** are similar in shape but may vary in size. As shown in FIGS. **40** and **41**, the discharge valve **138** is slightly larger than suction valve **214**.

[0266] Continuing with FIGS. **64-67**, the discharge valve **138** is shown in more detail. The suction valve **214** has the same features as the discharge valve **138** so only the discharge valve **138** is shown in more detail in the figures. The discharge valve **138** comprises a stem **402** joined to a body **428**. The body **428** comprises an outer rim **430** joined to a valve insert **432** by a tapered seating surface **434**. An annular cutout **436** formed within the seating surface **434** is configured to house a seal **438**, as shown in FIG. **67**.

[0267] Continuing with FIGS. **40** and **41**, during operation, the seating surface **434** and the seal **438** engage the seating surface **372** of the discharge surface **334** and block fluid from entering or exiting the discharge passages **360**, as shown in FIG. **40**. Likewise, the seating surface **434** and the seal **438** on the suction valve **214** engage the seating surface **366** of the suction surface **332** and block fluid from entering or exiting the suction passages **340**, as shown in FIG. **41**.

[0268] Continuing with FIGS. **40** and **41**, when the seating surfaces **434** and **372** are engaged, the valve insert **432** extends partially into the counterbore **380** formed in the discharge surface **334**. Fluid exiting the second openings **376** of the discharge passages **360** contacts the insert **432**, pushing the discharge valve **138** away from the discharge surface **334** before flowing around the seating surface **434** of the discharge valve **138**. Such motion enlarges the area for fluid to flow between the seating surfaces **372** and **434** before fluid reaches the surfaces **372** and **434**, thereby reducing the velocity of fluid flow within such area. The lowered fluid velocity between the surfaces **372** and **434** causes any wear to the valve **138** or **214** to be concentrated at the insert **432** instead of the crucial sealing elements, thereby extending the life of the valve **138** or **214**.

[0269] Likewise, the insert **432** on the suction valve **214** extends partially into the opening of the blind bore **342**. Fluid within the blind bore **342** contacts the insert **432** before flowing around the seating surface **434** and seal **438** of the suction valve **214**. Such motion enlarges the area for fluid to flow between the seating surfaces **366** and **434** before fluid reaches the surfaces **366** and **434**, thereby reducing the velocity of fluid flow within such area.

[0270] Continuing with FIGS. **64-67**, the stem **402** projects from a top surface **440** of the body **428** of the valve **138** or **214**. The outer rim **430** surrounds the stem **402** and is spaced from the stem **402** by an annular void **442**. A groove **444** is formed in the outer rim **430** for receiving a portion of a spring **446**, as shown in FIGS. **40** and **41**.

[0271] Continuing with FIGS. **40** and **41**, during operation, the valves **138** and **214** move axially along the longitudinal axis **62** of the housing **60** between open and closed positions. In the closed position, the seating surface **434** and the seal **438** of each of the valves **138** and **214** tightly engage the corresponding seating surface **372** or **366** of the fluid routing plug **132** and the valve insert **432** is disposed within the corresponding bore **380** or **342**. In the open position, the seating surface **434** and the seal **438** are spaced from the corresponding seating surface **372** or **366** of the fluid routing plug **132** and the valve insert **432** is spaced from the corresponding bore **380** or **342**.

Suction Valve Guide

[0272] With reference to FIGS. **40**, **41**, and **68-71**, axial movement of the suction valve **214** is guided by the suction valve guide **212**. The suction valve guide **212** comprises a thin-walled skirt **448** joined to a body **450** by a plurality of support arms **452**. The skirt **448** comprises a tapered upper section **454** joined to a cylindrical lower section **456**. The plurality of arms **452** join the tapered upper section **454** to the body **450**. A plurality of flow ports **458** are formed between adjacent arms **452** such that fluid may pass through the suction valve guide **212** during operation.

[0273] Continuing with FIGS. **40** and **41**, the suction valve guide **212** is installed within the housing **60** such that the tapered upper section **454** engages a tapered surface **455** of the walls surrounding the horizontal bore **70**. Such engagement prevents further axial movement of the suction valve guide **212** within the housing **60**. When the suction valve guide **212** is installed within the housing **60**, the skirt **448** covers the walls of the housing **60** positioned between the flow ports **458** and the fluid routing plug **132**. During operation, fluid wears against the skirt **448**, thereby protecting the housing **60** from wear and erosion. If the skirt **448** begins to erode, the suction valve guide **212** can be removed and replaced with a new guide **212**.

[0274] Continuing with FIGS. **40**, **41**, and **68-71**, the body **450** of the suction valve guide **212** is tubular and is centered within the skirt **448**. A tubular insert **460** is installed within the body **450**, as shown in FIG. **71**. The insert **460** is configured to receive the stem **402** of the suction valve **214**, as shown in FIGS. **40** and **41**. During operation, the stem **402** moves axially within the insert **460** and wears against the insert **460**. An annular cutout **462** formed in the stem **402**, shown in FIGS. **66** and **67**, provides space for any fluid or other material trapped between the stem **402** and the insert **460**. The insert **460** is made of a harder and more wear resistant material than the body **450** thereby extending the life of the suction valve guide **212**. For example, if the body **450** is made of stainless steel, the insert **460** may be made of tungsten carbide.

[0275] Continuing with FIGS. **40** and **41**, a spring **446** is positioned between the outer rim **430** of the suction valve **214** and the plurality of arms **452** such that the spring **446** surrounds at least a portion of the body **450** of the suction valve guide **212**. During operation, the spring **446** biases the suction valve **212** in a closed position, as shown in FIG. **41**. Fluid pushing against the valve insert **432** moves the suction valve **214** axially to compress the spring **446** and move the suction valve **214** to an open position, as shown in FIG. **40**.

Discharge Plug

[0276] With reference to FIGS. **40**, **41**, and **72-75**, axial movement of the discharge valve **138** is guided by the discharge plug **120**. The discharge plug **120** comprises a pair of legs **464** joined to a body **466**. The body **466** comprises a front portion **468** joined to a rear portion **470** by a medial portion **472**. The medial portion **472** has a larger outer diameter than both the front and rear portions **468** and **470**. An outer surface of the medial portion **472** engages the seal **126** installed within the first section **72** of the housing **60**, as shown in FIGS. **40** and **41**. The pair of legs **464** are joined to the medial portion **472** and extend between the medial portion **472** and the discharge surface **334** of the fluid routing plug **132**.

[0277] Continuing with FIGS. **40**, **41**, and **72-75**, a dowel opening **474** is formed in the outer surface of the medial portion **472** for receiving the locating dowel pin **130**. The discharge plug **120** is installed within the first section **72** of the housing **60** such that the locating dowel pin **130** is installed within the dowel opening **474** formed in the medial portion **472** and the locating cutout **128** formed in the first section **72** of the housing **60**. Such installation aligns the discharge plug **120** within the housing **60** so that the pair of legs **464** do not block the openings of the upper and lower discharge bores **102** and **104**.

[0278] Continuing with FIGS. **40** and **41**, the locating cutout **128** may be large enough to provide sufficient clearance for installation of the locating dowel pin **130** within the locating cutout **128**. The locating cutout **128** is sized to allow maximum clearance for assembly, but still maintain an acceptable rotational position of the discharge plug **120**. For example, the cutout **128** may be a maximum of 15 degrees wide along the circumference of the horizontal bore **70**.

[0279] Continuing with FIG. **75**, a blind bore **476** extends within the body **466** and opens on the rear portion **470** of the body **466**. The bore **476** is sized to receive a tubular insert **478**. The tubular insert **478** is similar to the tubular insert **460** installed within the suction valve guide **212**. The tubular insert **478** is configured to receive the stem **402** of the discharge valve **138**, as shown in FIGS. **40** and **41**.

[0280] Continuing with FIGS. **40**, **41**, and **75**, during operation, the stem **402** moves axially within the tubular insert **478**. A plurality of passages **480** are formed in the body **466** and interconnect the bore **476** and an outer surface of the medial portion **472**. During operation, any fluid or other material trapped within the bore **476** exits the discharge plug **120** through the passages **480**. A spring **446** is positioned between the medial portion **472** of the plug **120** and the outer rim **430** of the discharge valve **138**, as shown in FIGS. **40** and **41**. The spring **446** biases the discharge valve **138** in the closed position, as shown in FIG. **40**. Fluid pushing against the valve insert **432** moves the discharge valve **138** axially to compress the spring **446** and move the discharge valve **138** to an open position, as shown in FIG. **41**.

[0281] Continuing with FIGS. **40**, **41**, **72**, and **75**, the front portion **468** of the body **466** is sized to be disposed within a counterbore **482** formed within the front retainer **118**. When disposed therein, a rear surface **484** of the front retainer **118** abuts an outer surface of the medial portion **472** of the discharge plug **120**, as shown in FIGS. **40** and **42**. Such engagement holds the discharge plug **120** in place between the front retainer **118** and the fluid routing plug **132**. A blind bore **486** is formed in an outer surface of the front portion **468** of the plug **120**. The blind bore **486** is configured to engage a tool used to help install or remove the plug **120** from the housing **60**. For example, the bore **486** may have threaded walls.

Front Retainer

[0282] With reference to FIGS. **40**, **41**, **76**, and **77**, the front retainer **118** comprises opposed front and rear surfaces **488** and **484** joined by an outer surface having external threads **124** and a horizontal bore **490** formed therein. The horizontal bore **490** comprises a hex portion **492** that opens in the counterbore **482**, as shown in FIGS. **40** and **41**. The hex portion **492** is configured to mate with a tool used to thread the front retainer **118** into the housing **60** until it abuts the discharge plug **120**, as shown in FIGS. **40** and **41**. An annular void **494** is formed within the front surface **488** of the front retainer **118**. The annular void **494** decreases the weight of the front retainer **118**, making it easier to thread into the housing **60**.

Discharge Conduits and Manifold

[0283] With reference to FIG. **41**, each discharge fitting **108** comprises a support base **502** and a connection end **512**. A discharge fitting adapter **504** is installed within the counterbore **106** formed in the upper and lower discharge bores **102** and **104**. When installed, the seal **112** engages an outer surface of the fitting adapter **504**. A groove **505** is formed with the discharge fitting **108** for receiving a second seal **507**. The second seal **507** likewise engages an outer surface of the fitting adapter **504**.

[0284] Continuing with FIG. **41**, the support base **502** is sized to abut the outer intermediate surface **68** of the first section **72** of the housing **60**. The support base **502** comprises a plurality of passages **506**, shown in FIG. **29**, configured to align with the threaded openings **114** formed in the intermediate surface **68** and surrounding the discharge bores **102** and **104**. The threaded fasteners **116** are installed within the aligned passages **506** and openings **114** and tightened to secure the discharge fitting **108** to the first section **72**.

[0285] With reference to FIGS. **3** and **41**, the connection end **512** of the discharge fitting **108** is configured to mate within one or more discharge conduits **500** included in an upper or lower discharge manifold **514** or **516**, as shown in FIG. **3**. The upper and lower discharge manifolds **514** and **516** are supported on rack **518**, as shown in FIG. **3**. The fluid end assembly **52** is disposed within the interior open area of the rack **518**. The rack **518** supports the upper and lower discharge manifolds **514** and **516** in a spaced position from the discharge bores **102** and **104**. As a result, each discharge conduit **500** has an angled or bent shape. In operation, fluid discharges from the housing **60** through upper and lower discharge bores **102** and **104** and is carried to the corresponding upper or lower discharge manifolds **514** or **516** by the discharge fittings and conduits **108** and **500**.

Suction Conduits and Manifold

[0286] With reference to FIGS. **41** and **78**, each suction conduit **166** comprises the first connection member **164** joined to a second connection member **520** by threads, as shown in FIG. **78**. The first and second connection members **164** and **520** may be made of a metal or hardened material.

[0287] Continuing with FIG. **78**, the first connection member **164** comprises upper portion **524** joined to a lower portion **526**. External threads **172** are formed on a portion of the lower portion **526** for mating with the internal threads **170** formed in the suction bores **160** or **162**. The seal **176** installed within the housing **60** engages a cylindrical outer surface of the lower portion **526** below the external threads **172**. The upper portion **524** has a larger outer diameter than the lower portion **526** and is positioned outside of the housing **60**. The lower portion **526** abuts the counterbore **168** of the suction bores **160** and **162** of the second section **74** of the housing **60**.

[0288] With reference to FIGS. **3** and **78**, the second connection member **520** is configured to mate with one or more connection members or hoses **528** formed on an upper or lower suction manifold **530** or **532**. The upper and lower suction manifolds **530** and **532** are supported on the rack **518** adjacent the discharge manifolds **514** and **516**. The connection members or hoses **528** may be flexible so that they may bend, as needed, to properly interconnect the suction conduits **166** and the suction manifolds **530** and **532**. In operation, fluid is drawn into the housing **60** from the suction manifolds **530** and **532** via the connection members **528**, the suction conduits **166**, and the upper and lower suction bores **160** and **162**.

Assembly of Fluid End Section and Assembly

[0289] Turning to FIGS. **9**, **29**, **79**, and **80**, prior to assembling the housing **60**, the wear ring **136** is preferably first pressed into the counterbore **134** formed in the first section **72** of the housing **60**. Likewise, the hardened insert **210** is pressed into the counterbore **208** formed in the third section **76** of the housing **60**. The seals **126**, **112**, and **176** may also be installed within the first and second sections **72** and **74** of the housing **60**. The wear ring **222** and seal **224** may also be installed within the third section **76** of the housing **60** prior to assembling the housing **60**.

[0290] Following installation of the above described components, the housing **60** may be assembled as described above. Thereafter, the retention plates **206**, stuffing box **198**, rear retainer **200**, plunger packing **300**, and packing nut **290** may be attached to the rear surface **66** of the housing **60**. The inner components of the housing **60** are inserted within the housing **60** through the front surface **64** of the first section **72**. The inner components may be installed prior to attaching the components to the rear surface **66** of the housing **60**, if desired. Following assembly of each fluid end section **56**, each section **56** is attached to the power end assembly **54** using the stay rods **58**.

[0291] Each fluid end section **56** and its various components are heavy and cumbersome. Various tools or lifting mechanisms may be used to assemble the fluid end assembly **52** and attach it to the power end assembly **54**, creating the high pressure pump **50**.

Operation of Fluid End Assembly

[0292] Turning back to FIGS. **40** and **41**, in operation, retraction of the plunger **216** out of the housing **60** pulls fluid from the upper and lower suction bores **160** and **162** into the suction passages **340** within the fluid routing plug **132**. Fluid flowing through the suction passages **340** and into the blind bore **342** pushes on the valve insert **432** of the suction valve **214**, causing the valve **214** to compress the spring **446** and move to an open position, as shown in FIG. **40**. When in the open position, fluid flows around the suction valve **214** and the suction valve guide **212** and into the open horizontal bore **70** within the third section **76** of the housing **60**.

[0293] Continuing with FIG. **41**, extension of the plunger **216** further into the housing **60** pushes against fluid within the open horizontal bore **70** and forces the fluid towards the suction surface **332** of the fluid routing plug **132**. Such motion also causes the suction valve **214** to move to a closed position, sealing the opening of the blind bore **342**. Because the bore **342** is sealed, fluid is forced into the discharge passages **360**.

[0294] Fluid flowing through the discharge passages **360** contacts the valve insert **432** on the discharge valve **138**, causing the discharge valve **138** to compress the spring **446** and move into an open position, as shown in FIG. **41**. When in the open position, fluid flows around the discharge valve **138** and into the upper and lower discharge bores **102** and **104**. Because fluid exiting the discharge passages **360** has been compressed by extension of the plunger **216** into the housing **60**, such fluid has a higher fluid pressure than that entering the housing **60** through the suction bores **160** and **162**.

[0295] During operation, the plunger **216** continually reciprocates within the housing **60**, pressurizing all fluid drawn into the housing **60** through the suction bores **160** and **162**. Pressurized fluid exiting the housing **60** through the upper and lower discharge bores **102** and **104** is delivered to the upper and lower discharge manifolds **514** and **516** in communication with each of the fluid end sections **56**. Pressurized fluid within the discharge manifolds **514** and **516** is eventually delivered to the wellhead **18**, as shown in FIG. **2**.

ALTERNATIVE EMBODIMENTS

[0296] Turning to FIGS. **81-119**, alternative embodiments of fluid routing plugs that may be used with the fluid end assembly **52** are described in more detail. The alternative embodiments of fluid routing plugs are shown installed within alternative embodiments of fluid end sections in FIGS. **81-119**. Such fluid end section embodiments are described in more detail in U.S. Pat. No. 11,808,254 authored by Thomas, et al., and filed on Aug. 10, 2022, the entire contents of which are incorporated herein by reference. To the extent any features of the alternative fluid end section embodiments are the same or nearly the same as features of the fluid end section **56**, for ease of reference, such features will be given the same reference numbers as those described above. Likewise, to the extent any features of the fluid routing plug **132** and the various fluid routing plug embodiments are the same or nearly the same, such features will be given the same reference numbers herein, for ease of reference.

[0297] With reference to FIGS. **81-100**, another embodiment of a fluid routing plug **600** is shown. The fluid routing plug **600** is installed within another embodiment of a housing **602**. The housing **602** comprises an integral first and second section **604** joined to the third section **76**, as shown in FIG. **81**. The fluid routing plug **600** comprises a body **606** having a suction surface **608** and an opposed discharge surface **610** joined by an outer intermediate surface **612**, as shown in FIGS. **84-87**. The suction and discharge surfaces **608** and **610** are generally identical to the suction and discharge surfaces **332** and **334** of the fluid routing plug **132**.

[0298] Continuing with FIGS. **81-100**, a central longitudinal axis **613** extends through the body **606** and the suction and discharge surfaces **608** and **610**, as shown in FIG. **87**. When the fluid routing plug **600** is installed within the housing **602**, most of the plug **600** is installed within the integral section **604**, but at least a portion of the suction surface **608** is positioned within the third section **76** of the housing **602**, as shown in FIG. **81**.

[0299] Continuing with FIGS. **84-100**, the body **606** further comprises a plurality of suction fluid passages **614**. The suction passages **614** interconnect the intermediate surface **612** and the suction surface **608** of the body **606**, as shown in FIG. **87**. The connection is formed within an axially-blind bore **616** formed within the suction surface **608** of the body **606**. The suction passages **614** are generally identical to the suction passages **340** formed in the fluid routing plug **132**, but the passages **614** are oriented differently within the body **606**.

[0300] Continuing with FIGS. **88** and **89**, an opening **618** of each suction passage **614** on the intermediate surface **612** comprises a first side wall **620** joined to a second side wall **622** by first and second ends **624** and **626**. The first and second side walls **620** and **622** are straight lines of equal length S**1**, and the first and second ends **624** and **626** are circular arcs, as shown in FIG. **89**.

[0301] Continuing with FIG. **89**, the first end **624** of the opening **618** has a radius of R**1** with a center at C**1**, and the second end **626** has a radius of R**2** with a center at C**2**. The first end **624** is larger than the second end **626** such that R**1**>R**2**. The first and second side walls **620** and **622** are tangent to the first and second ends **624** and **626** and have an included angle, σ**1**. The opening **618** is oriented such that the center C**1** of the first end **624** is a perpendicular distance D from the central longitudinal axis **613**.

[0302] Continuing with FIG. **89**, the opening **618** has a centerline **628** that connects the centers C**1** and C**2** of the first and second ends **624** and **626**. The centerline **628** has a length E**1** and forms an angle χ with the central longitudinal axis **613**, further orienting the opening **618**. A cross-sectional shape of each suction passage **614** throughout the length of the body **606** corresponds with the shape of each opening **618**, as shown in FIGS. **84** and **87**. Each suction passage **614** is sized and shaped to maximize fluid flow through the passage **614** and minimize fluid turbulence and stress to the body **606** of the fluid routing plug **600**.

[0303] With reference to FIGS. **91** and **97**, each suction fluid passage **614** extends between the axially-blind bore **616** and the intermediate surface **612** such that each suction passage **614** comprises a first longitudinal axis **630** and a second longitudinal axis **632**. The first longitudinal axis **630** extends through the center C**1** of the first end **624** of the opening **618**, as shown in FIG. **91**. Like the center C**1**, the first longitudinal axis **630** is offset a perpendicular distance D from the central longitudinal axis **613**. The first longitudinal axis **630** does not intersect, is not parallel to, and is not co-planar with the central longitudinal axis **613**.

[0304] The second longitudinal axis **632** extends through point P, shown on FIG. **91**. The second longitudinal axis **632** may intersect and be co-planar with the central longitudinal axis **613**, as shown in FIGS. **91** and **97**. The configuration of the suction passages **614** encourages a vortex type fluid flow about the central longitudinal axis **613** of the body **606**, thereby reducing fluid turbulence and erosion during operation.

[0305] Continuing with FIGS. **85-87**, the body **606** further comprises a plurality of discharge fluid passages **634**. The discharge passages **634** interconnect the suction surface **608** and the discharge surface **610** of the body **606** and do not intersect any of the suction passages **614**. The discharge passages **634** are generally identical to the discharge passages **360** formed in the fluid routing plug **132**, but the passages **634** are oriented differently within the body **606**.

[0306] With reference to FIGS. **85**, **86**, **90**, **92**, and **93**, each discharge passage **634** opens at a first opening **636** on an outer rim **638** of the suction surface **608**, as shown in FIGS. **92** and **93**, and opens at a second opening **640** on a central base **642** of the discharge surface **610**, as shown in FIGS. **85** and **86**. A position of the first and second openings **636** and **640** of each discharge passage **634** may be determined relative to a plane containing a line **644** that is perpendicular to the central longitudinal axis **613**, as shown in FIG. **90**. The first opening **636**, when projected onto the plane, is positioned at a first distance F**1** from the central longitudinal axis **613** and at a first angle φ**1** relative to the line **644**. The second opening **640**, when projected onto the plane, is positioned at a second distance F**2** from the central longitudinal axis **613** and at a second angle φ**2** relative to the line **644**.

[0307] The first and second distances F**1** and F**2** shown in FIG. **90** are different. Likewise, the first and second angles φ**1** and φ**2** shown in FIG. **90** are different. In alternative embodiments, the first and second angles φ**1** and φ**2** may be different. In further alternative embodiments, the first and second angles φ**1** and φ**2** may be the same, but the first and second distances F**1** and F**2** may be different. In even further alternative embodiments, the first and second distances F**1** and F**2** may be the same, and the first and second angles φ**1** and φ**2** may be the same.

[0308] With reference to FIGS. **94-100**, each discharge passage **634** has an arced cross-sectional shape. The length of the arc may gradually increase between the suction and discharge surfaces **608** and **610**, as shown in FIGS. **94-96**. At least a portion of each discharge passage **634** intersects a plane containing the central longitudinal axis **613** of the body **606**, as shown in FIG. **97**.

[0309] Continuing with FIGS. **97-100**, the configuration of the discharge passages **634** encourages a vortex type flow of fluid about the central longitudinal axis **613** of the body **606**, thereby reducing fluid turbulence during operation. The shape of each discharge passage **634** thus maximizes fluid flow and minimizes stress to the body **606** of the fluid routing plug **600** during operation.

[0310] Turning back to FIG. **87**, the intermediate surface **612** comprises a first sealing surface **650** positioned adjacent the discharge surface **610** and a second sealing surface **652** positioned adjacent the suction surface **608**. The first and second sealing surfaces **650** and **652** each extend around the entire intermediate surface **612** in an endless manner and surround the longitudinal axis **613** of the body **606**.

[0311] Turning back to FIG. **81**, the first sealing surface **650** engages a first seal **654** installed within a groove **656** formed in the integral section **604** of the housing **602**. The second sealing surface **652** engages a second seal **660** installed within the counterbore **208** formed in the third section **76** of the housing **602**. The second seal **660** is surrounded by a wear ring **662**. Engagement of the first and second sealing surfaces **650** and **652** with the first and second seals **654** and **660** prevents fluid from leaking around the fluid routing plug **600** during operation. Because the first and second seals **654** and **660** are installed within the housing **602**, no grooves are formed in the fluid routing plug **600** for housing a seal.

[0312] Continuing with FIG. **87**, the intermediate surface **612** of the fluid routing plug **600** further comprises a first bevel **664** and a second bevel **666**. The first bevel **664** is positioned between the first sealing surface **650** and the suction passages **614**, and the second bevel **666** is positioned between the suction passages **614** and the second sealing surface **652**. The bevels **664** and **666** are configured to engage the first and second beveled surfaces **668** and **670** formed in the walls of the integral section **604** of the housing **602**, as shown in FIGS. **82** and **83**. Axial movement of the fluid routing plug **600** towards the stuffing box **198** is prevented by engagement of the first and second bevels **664** and **666** with the first and second beveled surfaces **668** and **670**, as shown in FIG. **81**.

[0313] Continuing with FIG. **82**, when the fluid routing plug **600** is installed within the housing **602**, the first bevel **664** seats against the first beveled surface **668**. The bevel **664** and the beveled surface **668** meet at a non-right angle. Such angle reduces stress in the fluid routing plug **600** and the housing **602** during operation. The bevel **664** and the beveled surface **668** remain engaged during the forward and backwards stroke of the plunger **216**.

[0314] Turning to FIG. **83**, in contrast to the first bevel **664**, the second bevel **666** is sized to be spaced from the second beveled surface **670** when the fluid routing plug **600** is initially installed within the housing **602**, as shown by a gap **672**. The gap **672** provides space for the fluid routing plug **600** to expand during operation.

[0315] As the plunger **216** retracts backwards away from the housing **602**, a significant amount of load is applied to the first bevel **664**. The applied load causes the fluid routing plug **600** and the integral section **604** to deform, allowing the second bevel **666** to eventually engage the second beveled surface **670**. Upon engaging the second beveled surface **670**, the load being applied to the first bevel **664** is shared with the second bevel **666**, thereby decreasing the load applied to the first bevel **664**. Without the gap **672**, the fluid routing plug **600** may not have enough room to deform, potentially causing damage to the fluid routing plug **600** and the housing **602** over time.

[0316] As the plunger **216** extends forward into the housing **602**, the second bevel **666** will return to its non-deformed state, re-creating the gap **672**. The gap **672** will repeatedly be created and closed during operation as the plunger **216** reciprocates. In addition to providing space for the fluid routing plug **600** to deform, the gap **672** also provides a gas and fluid relief area during the forward stroke of the plunger **216**.

[0317] Continuing with FIG. **82**, the intermediate surface **612** of the fluid routing plug **600** further comprises a transition bevel **674**. The transition bevel **674** extends between the first sealing surface **650** and the first bevel **664**, as shown in FIG. **87**. The transition bevel **674** does not engage the first beveled surface **668**. The transition bevel **674** helps reduce friction between the fluid routing plug **600** and the housing **602** during installation.

[0318] Turning to FIG. **101**, another embodiment of a housing **682** having the fluid routing plug **600** installed therein is shown. The housing **682** is identical to the housing **602**, but the integral section **604** has been split into a first section **684** joined to a second section **686** by a plurality of fasteners (not shown). Splitting the integral section **604** into two sections allows a wear ring **688** to be installed within a counterbore **690** formed in the first section **684**. The first seal **654** is also installed within the counterbore **690**. The wear ring **668** surrounds the first seal **654** and provides a barrier between the walls of the housing **682** and the first seal **654**, during operation. The wear ring **688** and the first seal **654** are held within the counterbore **690** by joining the second section **686** to the first section **684** of the housing **682**. During operation, if either the first seal **654** or the wear ring **688** begins to wear and erode, the first seal **654** and/or the wear ring **688** can be removed and replaced with a new first seal **654** and/or wear ring **688**. Using the wear ring **688** helps prevent wear or damage to the housing **682** during operation.

[0319] Turning to FIGS. **102** and **103**, another embodiment of a fluid routing plug **700** is shown. The fluid routing plug **700** is shown installed within the housing **682**. The fluid routing plug **700** is identical to the fluid routing plug **600**, but it comprises another embodiment of an intermediate surface **702**. Instead of a first sealing surface **650**, the intermediate surface **702** comprises a first groove **704** for housing the first seal **654**. Thus, the first seal **654** is installed within the fluid routing plug **700** instead of the counterbore **690**. The counterbore **690** only houses the wear ring **688**. Thus, the counterbore **690** may vary in size depending on components to be installed therein. The first seal **654** engages the wear ring **688** installed within the counterbore **690**. During operation, if either the first seal **654** or the wear ring **688** begins to wear and erode, the first seal **654** and/or the wear ring **688** can be removed and replaced with a new first seal **654** and/or wear ring **688**.

[0320] Turning to FIGS. **104** and **105**, another embodiment of a fluid routing plug **750** is shown installed within the housing **602**. The fluid routing plug **750** is identical to the fluid routing plug **600**, but it comprises another embodiment of an intermediate surface **752**. Instead of a second sealing surface **652**, the intermediate surface **752** comprises a second groove **754** for housing the second seal **660**. Thus, the second seal **660** is installed within the fluid routing plug **750** instead of the counterbore **208** of the third section **76**. Only the wear ring **662** is installed within the counterbore **208**. The second seal **660** engages the wear ring **662**. During operation, if either the second seal **660** or the wear ring **662** begins to wear and erode, the second seal **660** and/or the wear ring **662** can be removed and replaced with a new second seal **660** and/or wear ring **662**. In alternative embodiments, a fluid routing plug may have both the first and second seals **654** and **660** installed within grooves formed in the fluid routing plug.

[0321] Turning to FIGS. **106-112**, another embodiment of a fluid routing plug **800** is shown. The fluid routing plug **800** is identical to the fluid routing plug **600**, but it comprises another embodiment of an intermediate surface **802**. The fluid routing plug **800** is shown installed within another embodiment of a housing **804**. The housing **804** is identical to the housing **60**, but it includes another embodiment of a second section **806**. An outer intermediate surface of the second section **806** is shaped like the second section **686**, shown in FIG. **100**, but the interior walls of the second section **806** are modified, as described below.

[0322] Continuing with FIGS. **107-112**, instead of the first bevel **664**, the intermediate surface **802** of the fluid routing plug **800** comprises a cylindrical section **808** joined to a transition section **810** by a landing bevel **812**, as shown in FIG. **112**. The cylindrical section **808** and the landing bevel **812** are configured to engage a landing bevel counterbore **814** formed in the second section **806** of the housing **804**, as shown in FIG. **111**.

[0323] Continuing with FIG. **111**, the landing bevel **812** is configured to only contact the landing bevel counterbore **814** at a single point or very small engagement length. The cross-sectional profile of the landing bevel **812** is thus characterized as a splined curve. In operation, the varying fluid pressures in and around the fluid routing plug **800** cause the plug **800** to stretch or deform as the plunger **216** reciprocates. The landing bevel **812** and landing bevel counterbore **814** are sized so that the surfaces more fully engage as the fluid routing plug **800** stretches or deforms during operation. Such engagement supports the housing **804** and fluid routing plug **800** when in their highest state of loading, thereby reducing stress and increasing the life span of such components.

[0324] Continuing with FIGS. **106-110**, the intermediate surface **802** further comprises the first groove **704** for housing the first seal **654**, as shown in FIG. **111**, and comprises the second sealing surface **652**. However, the fluid routing plug **800** does not comprise the second bevel **666**. The intermediate surface **802** merely tapers outwardly from the transition section **810** to the second sealing surface **652**. The intermediate surface **802** between the suction passages **314** and the second sealing surface **652** is sized to closely face the walls of the second section **806** of the housing **804**. Axial movement of the fluid routing plug **800** within the housing **804** is prevented by engagement of the landing bevel **812** and the landing bevel counterbore **814**, as shown in FIG. **111**.

[0325] Turning to FIGS. **113-119**, another embodiment of a fluid routing plug **900** is shown. The fluid routing plug **900** is identical to the fluid routing plug **800**, but it comprises another embodiment of an intermediate surface **902**. The intermediate surface **902** is like the intermediate surface **802**, with a few modifications. The fluid routing plug **900** is shown installed within another embodiment of a housing **904**. The housing **904** is identical to the housing **60**, but it comprises another embodiment of a second section **906**.

[0326] Continuing with FIGS. **114-119**, the intermediate surface **902** comprises a cylindrical section **908** joined to a transition section **910** by a landing bevel **912**. In contrast to the landing bevel **812**, the landing bevel **912** joins the cylindrical section **908** at a sharper angle. However, at least a portion of the landing bevel **912** is still a splined curve. A relief **914** is also formed in the walls of the housing **904** adjacent a landing bevel counterbore **916**. The relief **914** allows the landing bevel counterbore **916** to be formed in the walls of the housing **904** at a sharper angle. The shape of the landing bevel counterbore **916** is a straight bevel. The modified configuration of the landing bevel **912** and the landing bevel counterbore **916** allows the surfaces to more fully engage during operation. Specifically, more surface area of the landing bevel **912** engages more surface area of the landing bevel counterbore **916** as the fluid routing plug **900** stretches and deforms during operation. Distributing the load during operation over a larger surface area reduces the stresses within the housing **904** and the fluid routing plug **900**.

[0327] Continuing with FIGS. **114-119**, a first groove **918** for housing a first seal **920** is formed within the cylindrical section **908**. In contrast to the first groove **704**, the first groove **918** extends a greater length of the cylindrical section **908** and houses a larger first seal **920**. By increasing the size of the seal **920** installed within the fluid routing plug **900**, the area of the intermediate surface **902** contacted by high pressure fluid is reduced, thereby reducing the total stress applied to the fluid routing plug **900** during operation.

[0328] Continuing with FIGS. **114-119**, the transition section **910** of the intermediate surface **902** has a smaller outer diameter than that of the fluid routing plug **800**. The smaller diameter increases the size of the annular fluid channel **412** formed between the fluid routing plug **900** and the walls of the housing **904** surrounding the suction bores **160** and **162**, as shown in FIG. **113**. The walls of the housing **904** surrounding the suction bores **160** and **162** may include one or more bevels **922** to further increase the size of the annular fluid channel **412**. The increase in size of the annular fluid channel **412** helps optimize fluid flow within the housing **904**.

[0329] Continuing with FIGS. **117** and **119**, to account for the smaller diameter of the transition section **910**, the intermediate surface **902** further comprises a step **924** positioned between the suction passages **614** and the second sealing surface **652**, as shown in FIG. **117**. The step **924** increases the outer diameter of the intermediate surface **902** so that the second sealing surface **652** engages the second seal **660**, as shown in FIG. **113**.

[0330] Turning now to FIGS. **120-137**, another embodiment of a fluid routing plug **1000** is shown. Fluid routing plug **1000** extends the suction seat **1012** and discharge seat **1027** radially compared to those in embodiments **132**, **600**, **700**, **750**, **800**, and **900** described herein, without requiring any changes to the relevant fluid end sections. Multi-sectioned discharge fluid passages **1005** are utilized to prevent partial blockage of these passages due to the radial extent of the suction seat **1012**. This and other features are discussed in more detail hereinafter.

[0331] Fluid routing plug **1000** has a generally cylindrical shape and comprises suction surface **1001**, discharge surface **1002**, outer intermediate surface **1003**, a plurality of suction fluid passages **1004**, a plurality of discharge fluid passages **1005**, a front transition radius **1006**, a rear transition radius **1007**, a central longitudinal axis **1008**, and a transverse axis **1009**. To align with the directional convention used throughout the disclosure of fluid routing plug **1000** the suction surface **1001** may be referred to as the rear surface **1001**, and the discharge surface **1002** may be referred to as the front surface **1002**.

[0332] The suction surface **1001**, shown in FIGS. **124** and **125**, has a circular projection with the central longitudinal axis **1008** as the center. Beginning at the outermost radius of the suction surface **1001** and moving toward the center, the suction surface **1001** comprises a plurality of concentric elements. These elements are the outer rim **1010**, suction crown **1011**, suction seat **1012**, first radius **1013**, suction bore **1014**, second radius **1015**, and suction base **1016**. The suction base **1016** comprises a circular boss **1018** concentric with the suction surface **1001**. The circular boss **1018** comprises third radius **1019**, first bevel **1020**, fourth radius **1021**, and boss surface **1022** which are also concentric with the suction surface **1001**.

[0333] The suction surface **1001** further comprises a plurality of locating bores **1017**, which are blind bores spaced evenly around the circumference. These locating bores **1017** are positioned radially at the section of the suction surface **1001** that extends the furthest longitudinally. Specifically, their radial placement falls within the radial limits of the suction crown **1011**.

[0334] During manufacturing, the suction surface **1001** may be placed on a planar surface to facilitate operations on other surfaces of the fluid routing plug **1000**. The planar surface may include projecting elements that fit tightly within the locating bores **1017**. The radial placement of the locating bores **1017** aligns with the contact area between the planar surface and the suction surface **1001**, minimizing the bending moment on the projecting elements. This reduces the deflection in the projecting elements and limits movement of the fluid routing plug **1000** during these operations.

[0335] The discharge surface **1002**, shown in FIGS. **122-123**, also has a circular projection centered on the central longitudinal axis **1008**. Beginning at the outermost radius of the discharge surface **1002** and moving toward the center, the discharge surface **1002** comprises a plurality of concentric elements. These elements are outer rim **1023**, fifth radius **1024**, outer crown **1025**, discharge crown **1026**, discharge seat **1027**, sixth radius **1028**, counterbore **1029**, seventh radius **1030**, and central base **1031**. The central base **1031** comprises installation bore **1033** which comprises blind bore **1034**, first chamfer **1035**, and base **1036** all of which are also concentric with the discharge surface **1002**. The blind bore **1034** may be threaded. In this embodiment the base **1036** is flat but may have other shapes. The transition from the bore **1034** to the base **1036** may be radiused to reduce stress concentration.

[0336] The discharge surface **1002** further comprises a plurality of locating bores **1032**, which are blind bores spaced evenly around the circumference. These locating bores **1032** are positioned radially at the section of the discharge surface **1002** that extends the furthest longitudinally. Specifically, their radial placement falls within the radial limits of the outer crown **1025** and discharge crown **1026**. Locating bores **1032** are used in the same manner as locating bores **1017** when manufacturing operations are performed on the surfaces other than the discharge surface **1002**.

[0337] The outer intermediate surface **1003**, shown in FIGS. **127** and **129-132**, comprises a plurality of sections with linear, radial, and spline profiles. Beginning at the section closest to the discharge surface **1002** and continuing along the outer intermediate surface **1003** to the section closest to the suction surface **1001**, the sections are: first cylindrical surface **1037**, first annular groove **1038**, second cylindrical surface **1039**, eighth radius **1040**, second bevel **1041**, ninth radius **1042**, third bevel **1043**, transition surface **1044**, suction passage surface **1045**, tenth radius **1046**, third cylindrical surface **1047**, eleventh radius **1048**, fourth bevel **1049**, twelfth radius **1050**, annular shoulder **1051**, stress relief cutout **1052**, fourth cylindrical surface **1053**, second annular groove **1054**, fifth cylindrical surface **1055**, thirteenth radius **1056**, and fifth bevel **1057**.

[0338] The first annular groove **1038**, as shown in FIG. **130**, comprises fourteenth radius **1058**, first sidewall **1059**, fifteenth radius **1060**, first base **1061**, sixteenth radius **1062**, second sidewall **1063**, and seventeenth radius **1064**. In this embodiment, the first annular groove **1038** is configured to receive a first seal **386** as shown in FIGS. **120** and **121**.

[0339] The stress relief cutout **1052**, shown in FIG. **127**, comprises eighteenth radius **1065**, linear section **1066**, and nineteenth radius **1067**. This and other embodiments of the stress relief cutout **1052** are disclosed in greater detail in U.S. patent application Ser. No. 18/933,692, which is co-pending herewith and incorporated by reference herein.

[0340] The second annular groove **1054**, shown in FIG. **132**, comprises twentieth radius **1068**, third sidewall **1069**, twenty-first radius **1070**, second base **1071**, twenty-second radius **1072**, fourth sidewall **1073**, and twenty-third radius **1074**. In this embodiment the second annular groove **1054** is configured to receive a second seal **1094** as shown in FIGS. **120-121**.

[0341] Each suction fluid passage **1004**, shown in FIGS. **126**, **129**, and **133**, fluidly connects the suction passage surface **1045** of the outer intermediate surface **1003** to the suction bore **1014** of the suction surface **1001**. The plurality of suction fluid passages **1004** are spaced evenly around the circumference of the fluid routing plug **1000** and do not intersect any of the discharge fluid passages **1005**. Each suction fluid passage **1004** comprises a longitudinal axis **1080**, along which the cross-section remains constant as shown in FIG. **133**. The cross-section of each suction fluid passage **1004** is generally oval or tear drop shaped. Specifically each suction fluid passage **1004** comprises a fifth sidewall **1075** and a sixth sidewall **1076**, which are joined by first and second curved surfaces **1077** and **1078**. The fifth and sixth sidewalls **1075** and **1076** are planar surfaces with widths S**75** and S**76**. In this embodiment S**75** equals S**76** but may differ in other embodiments.

[0342] Continuing with FIG. **133**, the first and second curved surfaces **1077** and **1078** are partial cylindrical surfaces. The first curved surface **1077** has a radius R**77** with a center at C**77**, while the second curved surface **1078** has radius R**78** with a center at C**78**. The fifth and sixth sidewalls **1075** and **1076** are tangent to the first and second curved surfaces **1077** and **1078**, forming an included angle, σ**4**. In this embodiment R**77** is greater than R**78**; however, in other embodiments, R**77** may be equal to or smaller than R**78**. Centers C**77** and C**78** lie on a centerline **1079** having a length E**79**. In this embodiment, centerline **1079** is parallel to the central longitudinal axis **1008** but may not be in other embodiments.

[0343] The suction fluid passage **1004** further comprises a twenty-fourth radius **1081** and a twenty-fifth radius **1082**, shown in FIG. **126**. These radii eliminate the sharp transitions between the fifth sidewall **1075**, sixth sidewall **1076**, first curved surface **1077**, and second curved surface **1078** to the suction passage surface **1045** at one end and the suction bore **1014** at the other end of the suction fluid passage **1004**.

[0344] The longitudinal axis **1080** of each suction fluid passage **1004** is colinear with center C**77** of the first curved surface **1077** and intersects the central longitudinal axis **1008** at angle Θ**4** as shown in FIG. **126**.

[0345] Each discharge fluid passage **1005** fluidly connects the suction surface **1001** to the discharge surface **1002**. The plurality of discharge fluid passages **1005** are spaced evenly around the circumference of the fluid routing plug **1000** and do not intersect any of the suction fluid passages **1004**. Each discharge fluid passage **1005** comprises a discharge surface section **1083**, a suction surface section **1084**, and a longitudinal axis **1085**. While the cross-section of the discharge surface section **1083** remains constant, the cross-section of the suction surface section **1084** varies along its length. Although the cross-section of the discharge fluid passage **1005** may change, the cross-sectional area remains constant throughout the entire length of the discharge fluid passage **1005**.

[0346] The discharge surface section **1083** begins at the discharge surface **1002** and terminates where it connects to the suction surface section **1084**. The cross-section of the discharge surface section **1083** is a curved slot with rounded ends as shown in FIG. **135**. The discharge surface section **1083** comprises plurality of partial cylindrical surfaces, including an inner wall **1086** and outer wall **1087**, which are connected tangentially to two end sections **1088**. The radius R**86** of the inner wall **1086** is smaller than the radius R**87** of the outer wall **1087**. Subtracting R**86** from R**87** gives the width W**83** of the discharge surface section **1083**. The circumferential length of each wall **1086** and **1087** is determined from the radii R**86** and R**87** and the swept angle Θ**83**.

[0347] The suction surface section **1084** begins at the termination of the discharge surface section **1083** and terminates at the suction surface **1001**. As stated earlier, the cross-section of the suction surface section **1084** varies along its length. At its beginning, the cross-section is the same as that of the discharge surface section **1083**. At its termination, the cross section is narrower and longer but maintains the same cross-sectional area as at the beginning.

[0348] The cross-section of the suction surface section **1084** at its termination, that is, at the suction surface **1001**, is shown in FIG. **137**. This configuration results in the suction surface section **1084** comprising a plurality of partial frustoconical surfaces, including an inner wall **1089** and outer wall **1090**, which are connected tangentially to two end sections **1091**. The radius R**89** of the inner wall **1089** is smaller than the radius R**90** of the outer wall **1090**. Subtracting R**89** from R**90** gives the width W**84** of the suction surface section **1084** at the suction surface **1001**. The circumferential length of each wall **1089** and **1090** at the suction surface **1001** is determined from the radii R**89** and R**90** and the swept angle Θ**84**. Note that W**84** is always smaller than W**83**, and Θ**84** is always larger than Θ**83**.

[0349] The suction surface section **1084** smoothly transitions from the from the cross-section at its beginning, which is the same cross section as the discharge surface section **1083**, to the cross-section at its termination. This is commonly referred to as a lofted surface. At any point along the suction surface section **1084** the cross-sectional area remains constant.

[0350] The center C**5** of the discharge fluid passage **1005** at any point can be obtained by subtracting the radius of the inner wall **1089** from the radius of the outer wall **1090**, dividing the result by two to determine the radius, and then dividing the swept angle by two to find the point along the circumference. The longitudinal axis **1085** forms an angle Θ**5** with the central longitudinal axis **1008**, as shown in FIG. **126**, and coincides with the center C**5** of the discharge fluid passage **1005** along its entire length.

[0351] During assembly of the fluid end section **56**, with the housing **60** already assembled and the insert **210** and wear ring **136** installed within the housing **60**, an externally threaded tool (not shown) is screwed into the installation bore **1033** and used to insert the fluid routing plug **1000** in the horizontal bore **70**. During the insertion process, the twelfth radius **1050**, fourth bevel **1049**, and eleventh radius **1048** assist with the initial alignment and insertion of the fluid routing plug **1000** through the wear ring **136**. As the fluid routing plug **1000** is inserted further into the horizontal bore **70**, the third bevel **1043**, ninth radius **1042**, second bevel **1041**, and eighth radius **1040** facilitate the final alignment prior to full insertion. Simultaneously, twelfth radius **1050**, fourth bevel **1049**, and eleventh radius **1048** also assist with the alignment and insertion of the fluid routing plug **1000** in the insert **210**. These combinations of radii and bevels not only assist in alignment but also prevent damage to the fluid routing plug **1000**, wear ring **136**, and insert **210** during the assembly process.

[0352] During operation, the seal **438** of the suction valve **214** never extends radially past the suction crown **1011** of the suction surface **1001** of the fluid routing plug **1000**. Similarly, the seal **438** of the discharge valve **138** never extends radially beyond the discharge crown **1026** of the discharge surface **1002** of the fluid routing plug **1000**. This positional limitation reduces the risk of overextension and tearing of the seals **438**.

[0353] The discharge fluid passages **1005** are never obstructed, even partially, by the seal **438** of the suction valve **214**. This is achieved by reducing the width W**84** of the suction surface section **1084** at the suction surface **1001**. Despite this reduction, fluid flow through the discharge fluid passage **1005** continues with minimal turbulence, as the cross-sectional area of the discharge fluid passage **1005** remains constant throughout its length.

[0354] The boss **1018** in the suction base **1016** of the suction surface **1001** increases the material thickness between the boss surface **1022** and the installation bore **1033** in the central base **1031** of the discharge surface **1002**. The additional material thickness reduces stress and lowers the likelihood of failure of the fluid routing plug **1000** in this area. The second radius **1015** of the suction surface **1001** and the seventh radius **1030** of the discharge surface **1002** also reduce stress concentrations in the same area again increasing expected life of the fluid routing plug **1000**.

[0355] To promote non-turbulent fluid flow, the design incorporates the twenty-fourth and twenty-fifth radii **1081** and **1082** of the suction fluid passage **1004**. Additionally, the first radius **1013** of the suction surface **1001** and the sixth radius **1028** of the discharge surface **1002** further refine the flow paths, reducing the potential for turbulence.

[0356] Referring now to FIG. **138**, another embodiment of a fluid routing plug **1100** is shown installed in another embodiment of a fluid end section **1156**. The fluid end section **1156** comprises a housing **1102** with a horizontal bore **1170** configured to receive an insert **11210**. The fluid routing plug **1100** comprises an annular shoulder **1151**. This illustrates another method for longitudinally positioning the fluid routing plug **1100**.

[0357] Referring now to FIGS. **120-121** and **139-141**, another embodiment of an insert **12210** is shown. In previous embodiments, the insert **210** experienced deformation due to the heavy press fit required for operation. Specifically, the front surface **416** displaced outward, forming a convex cone, while the rear surface **418** underwent plastic compression, resulting in a concave deformation. These deformations resulted in a point contact between the front surface **416** of the insert **210** and the annular shoulder **1051** of the fluid routing plug **1000**, as well as between the rear surface **418** of the insert **210** and the base **209** of the counterbore **208** of the dynamic body **76**, as shown in FIG. **139**. The resulting point contacts create stress concentrations within the insert **210** and fluid routing plug **1000** leading to premature failure of the insert **210** and or fluid routing plug **1000**.

[0358] Measurements of the resulting deformations indicate that the front and rear surfaces **416** and **418** remain planar, allowing the displacement to be quantified as a linear measure. The front surface **416** undergoes outward extrusion, with the displacement shown in FIG. **139** as **417**, while the rear surface **418** undergoes inward intrusion, with the displacement shown as **419**. The amount of the displacements is between 0.0004 and 0.0008 inches and they are not necessarily equal. The displacements may also be measured as an angular displacements. Note that the displacements in FIG. **139** are greatly exaggerated for illustrative purposes.

[0359] The solution to the deformation issue is a new embodiment of insert **12210**, shown in FIGS. **140** and **141**, which comprises a concave front surface **12416** and a convex rear surface **12418**. Insert **12210** is generally similar to insert **210**, shown in FIGS. **60-63**, except for the modified surfaces. FIGS. **140-141** illustrate these modifications by replacing insert **210** of FIG. **63** with insert **12210**. The front surface **12416** is pre-formed to include a specific displacement **12421**, and the rear surface **12418** is pre-formed with a corresponding displacement **12423**. This geometry is designed so that, when the insert is pressed into place, the concave front surface **12416** and convex rear surface **12418** account for the distortions caused by the heavy press fit, resulting in a flat final shapes of both surfaces **12416** and **12418** that are perpendicular to the longitudinal axis.

[0360] Once installed, as shown in FIG. **141**, the surfaces **12416** and **12418** are planar and perpendicular to the longitudinal axis of the insert **12210**, allowing for full or nearly full contact between the front surface **12416** of the insert **12210** and the annular shoulder **1051** of the fluid routing plug **1000**, as well as between the rear surface **12418** of the insert **12210** and the base **209** of the counterbore **208** of the dynamic body **76**. However, the amounts of the pre-formed displacements **12421** and **12423** do not necessarily match the measured displacements **417** and **419** after assembly of the original embodiment of insert **210**, as the modified surfaces of the new insert **12210** alter the amount of deformation due to the press fit. Additionally, this method of pre-forming the surfaces could also be applied in cases where the original displacements are not planar, adapting the insert geometry to achieve the desired final shape. Achieving the desired flatness may require a trial-and-error approach, with iterative adjustments to the geometry of insert **12210** to ensure optimal performance and minimize stress concentrations.

[0361] Referring now to FIGS. **142-148**, another solution to the deformation issue is shown. Unlike the previous approach, which modifies the insert **12210** itself, this solution modifies the surfaces that contact the deformed front and rear surfaces **416** and **418** of the insert **210** so that the surfaces conform to the deformation.

[0362] The surfaces modified **1351** and **13209**, are shown in a new embodiment of the fluid routing plug **1300** and a new embodiment of the dynamic body **1376**. The new embodiment of the fluid routing plug **1300** is shown in FIGS. **142-144**. The fluid routing plug **1300** comprises an annular shoulder **1351** having an offset of **13417**. The offset **13417** is shaped to match the displacement **417** of the front surface **416** of the insert **210** after installation. FIG. **144** illustrates the offset **13417** as an angular displacement, demonstrating an alternative way to quantify it.

[0363] The new embodiment of the dynamic body **1376** is shown in FIGS. **145-147**. The dynamic body **1376** comprises a counterbore **13208** with a seal groove **13211**, and a base **13209**, which has an offset **13419**. The offset **13419** is shaped to match the displacement **419** of the rear surface **418** of the insert **210** after installation. FIG. **147** illustrates the offset **13419** as an angular displacement, demonstrating an alternative way to quantify it.

[0364] Once assembled, the modified annular shoulder **1351** of the fluid routing plug **1300** and the base **13209** of the counterbore **13208** of the dynamic body **1376** fully, or nearly fully, engage the deformed front and rear surfaces **416** and **417** of the insert **210**, as shown in FIG. **148**. This engagement reduces stress concentrations in the insert **210**, which could otherwise lead to premature failure. As a result, this modification enhances the durability and service life of the assembly.

[0365] Unlike the previous solution, which may require a trial-and-error approach due to the modified shape of the insert **12210** affecting its deformation, this solution allows for post-insertion measurement of the insert **210**. The same deformations can then be replicated on the engaging surfaces **1351** and **13209** without the need for trial and error. Measurements may be taken on a non-modified dynamic body **76**, after which the insert **210** can be removed and the dynamic body **76** modified. Alternatively, a test configuration may be constructed to measure the displacement of both the front and rear surfaces **416** and **418** before modifying the surfaces **1351** and **13209**.

[0366] It is also contemplated that one or both of the deformations may not be planar, requiring a non-planar profile to be formed on the engaged surfaces **1351** and **13209**. Additionally, the deformations may vary for the front and rear surfaces **416** and **418**, resulting in different profiles on the annular shoulder **1351** and the base **13209** of the counterbore **13208**. In this embodiment, the deformations result in planar surfaces, which are addressed by modifications to the engaging surfaces **1351** and **13209**.

[0367] In operation, even with the load faces **1351** and **13209** modified to enhance contact surface area, the insert **210** still experiences variable loads about its circumference on the front and rear surfaces **416** and **418** due to the suction fluid passages **1304** in the fluid routing plug **1300**. The extremely high pressure applied to the fluid routing plug **1300** from the discharge surface **1302** is transferred to the insert **210** through the fluid routing plug **1300**. The suction fluid passages **1304** reduce the wall thickness between the discharge surface **1302** and the insert **210**. Specifically, the area of the fluid routing plug **1300** between the suction fluid passages **1304** and the insert **210** is less rigid than the sections between the suction fluid passages **1304**, allowing localized flexing of the fluid routing plug **1300**. This results in a lower force being applied to the insert **210** in those areas. The varying magnitude of force causes differing amounts of flexure in the insert **210**, creating areas of stress concentration and, consequently, regions more prone to early failure. The flexure is more pronounced if the counterbore **13208** of the dynamic body **1376** includes a seal groove **13211** as shown in FIG. **148**.

[0368] The solution to this problem of varying magnitudes of applied force and resulting flexure is to stiffen the fluid routing plug **1300**, significantly reducing or eliminating flexure in the section of the fluid routing plug **1300** between the suction fluid passages **1304** and the insert **210**. One such design is embodied in the fluid routing plug **1400** shown in FIGS. **149-150**. The fluid routing plug **1400** comprises a plurality of suction fluid passages **1404**, each incorporating a support column **1495**. The support column **1495** reinforces the area of the fluid routing plug **1400** between the insert **210** and the suction fluid passages **1404**, thereby reducing flexure as intended. To ensure that the inclusion of the support columns **1495** does not reduce flow capacity, the dimensions of the suction fluid passages **1404** are adjusted to maintain their original cross-sectional area.

[0369] Another embodiment of a fluid routing plug **1500** is shown in FIGS. **151-152**. The fluid routing plug **1500** comprises a plurality of suction fluid passages **1504**, each incorporating a support column **1595**. The support columns **1595** are wider than the support columns **1495** of fluid routing plug **1400**, providing increased stiffness to the area between the suction fluid passages **1504** and the insert **210**. This enhanced stiffness further reduces flexure. As with the previous embodiment, the dimensions of the suction fluid passages **1504** are adjusted to maintain their original cross-sectional area, ensuring that the inclusion of the support columns **1595** does not restrict flow.

[0370] It is acknowledged that the support columns **1495** and **1595** may, at first glance, appear to be nothing more than the spaces between a greater number of smaller suction fluid passages **1404** and **1504**, essentially halving the size of each passage **1404** and **1504** and doubling their number. However, the placement of these passages **1404** and **1504** cannot be arbitrarily placed due to the need to avoid interference with the discharge fluid passages **1405** and **1505**. Unlike a simple geometric redistribution, the arrangement of the suction fluid passages **1404** and **1504** and support columns **1495** and **1595** is carefully designed to balance structural integrity with fluid routing constraints. The result is a functional optimization that ensures stiffness is maximized without disrupting the required flow paths.

[0371] Another embodiment of a fluid routing plug **1600** is shown in FIGS. **153-161**. The fluid routing plug **1600** comprises a suction surface **1601** having a suction bore **1614**, an outer intermediate surface **1603**, a plurality of discharge passages **1605**, and a plurality of suction fluid passages **1604**. Each suction fluid passage **1604** has teardrop shape when viewed perpendicular to its longitudinal axis, with a larger end **1677** and a smaller end **1678**, as shown in FIG. **155**. In previous embodiments the larger end **1077** of the teardrop shape is oriented closer to the insert **210** upon assembly, as can be inferred from FIGS. **120**, **129**, and **133**. However, in fluid routing plug **1600**, this orientation is reversed at the outer intermediate surface **1603**, positioning the smaller end **1678** of the teardrop shape closer to the insert **210** when assembled.

[0372] Due to the angles of the discharge fluid passages **1605**, the orientation of the teardrop shape must transition back to having the larger end **1677** facing the insert **210** as the suction fluid passages **1604** approach the suction bore **1614**. Without this transition, the suction fluid passages **1604** would intersect the discharge fluid passages **1605**. This transition occurs gradually as the suction fluid passages **1604** progress from the outer intermediate surface **1603** to the suction bore **1614**. At the longitudinal midpoint of each suction fluid passage **1604** the cross-section is oval. FIGS. **156-161** illustrate the progression of this transformation from the outer intermediate surface **1603** to the suction bore **1614**. This configuration maximizes stiffness in areas where the thickness between the suction fluid passage **1604** and the insert **210** is smallest while maintaining the required clearances closer to the suction bore **1614**.

[0373] Another embodiment of a fluid routing plug **1700** is shown in FIGS. **162-165**. The fluid routing plug **1700** comprises an outer intermediate surface **1703** and plurality of suction fluid passages **1704**. The outer intermediate **1703** surface comprises a suction passage surface **1745**, a third cylindrical surface **1747**, an annular shoulder **1751**, and a fourth cylindrical surface **1753**. In this embodiment, the outside diameter of the suction passage surface **1745** is reduced to be smaller than the inside diameter of the insert **210**. Specifically, the diameter of the suction passage surface **1745** is smaller than the fourth cylindrical surface **1753**, while the third cylindrical surface **1747** retains its original diameter.

[0374] Since there is no longer any portion of the suction fluid passages **1704** behind the annular shoulder **1751**, the thickness of the third cylindrical surface **1747** is constant. As a result, while deflection may increase as compared to embodiments with material behind the annular shoulder **1751**, it does not vary around its circumference, which is beneficial for the life of the insert **210**. The suction fluid passages **1704** may or may not undergo transformation, as in embodiment **1600**.

[0375] The fluid routing plugs and corresponding fluid end sections described herein have various embodiments of fluid passages, intermediate surfaces, and corresponding housing sections. While not specifically shown in a figure herein, various features from one fluid routing plug or fluid end section embodiment may be included in another fluid end section embodiment.
[0376] Even after extensive efforts to distribute load evenly across the engaging faces of the insert and the fluid routing plug, extreme operational forces continue to cause localized deformation at the contact surfaces. Repeated qualitative and quantitative analyses have revealed that the most durable interface is achieved by initiating contact at the approximate center of the front face of the insert. As the applied load increases, this contact zone should expand both radially inward and outward until full engagement is achieved. This contact progression has been shown to significantly extend insert life under cyclic high-pressure operation by reducing stress concentrations and delaying plastic deformation. To implement this contact progression, the annular shoulder of the fluid routing plug is configured such that the portion of the annular shoulder opposite the center of the insert's front face makes initial contact. This principle of initial central contact can be realized through numerous shoulder profiles. Three embodiments of shoulder geometries developed by the inventors are described in the following paragraphs.
[0377] The first of the three embodiments, shown in FIGS. **166-172**, has a fluid routing plug **1800** with an annular shoulder **1851**. The annular shoulder **1851** comprises a flat middle section **1892** flanked by an inner taper **1897** and an outer taper **1899**. The flat middle section **1892** may be radially centered on the front face **18416** of the insert **18210**, may extend generally perpendicular to the longitudinal axis **1808**, and may extend from a first radius **18101** to a second radius **18103**. The flat middle section **1892** is spaced longitudinally from the linear section **1866** of the stress relief cutout **1852** and that spacing defines the height **18105** of the flat middle section **1892**. The inner taper **1897** extends from the first radius **18101**, at the termination of the flat middle section **1892**, to the linear section **1866**, transitioning into the linear section **1866** with an eighteenth radius **1865**. Radially outward from the second radius **18103**, the flat middle section **1892** transitions into the outer taper **1899**, which extends from the second radius **18103** to the outer intermediate surface **1803**, terminating at the same longitudinal position as the linear section **1866**. However, the outer taper **1899** is truncated by the twelfth radius **1850** at the outer intermediate surface **1803**.
[0378] In one example embodiment, the height **18105** of the flat middle section **1892** is 0.0003 inches, the first radius **18101** located 0.100 inches radially outward from the termination of the linear section **1866**, and the second radius **18103** located 0.060 inches radially inward from the outer intermediate surface **1803**. These specific dimensional relationships are provided for illustrative purposes and may be varied based on application-specific requirements. One or both of the inner and outer tapers **1897** and **1899** may comprise multiple linear segments having differing taper angles, may be curved segments with a constant radius, or may be spline curves. Additionally, the inner and outer tapers **1897** and **1899** may have equal or unequal lengths, depending on the desired progression of contact during loading.
[0379] The second of the three embodiments, shown in FIGS. **173-174**, has a fluid routing plug **1900** with an annular shoulder **1951**. The annular shoulder **1951** comprises a single symmetric splined curve having a crown **1992** at the point the annular shoulder **1951** contacts the front surface **19416** of the insert **19210**. The crown **1992** is centered on the annular shoulder **1951**. The crown **1992** is spaced longitudinally from the linear section **1966** of the stress relief cutout **1952**, and that spacing defines the height **19105** of the crown **1992**. The annular shoulder **1951** transitions to the linear section **1966** with an eighteenth radius **1965** and is truncated by the twelfth radius **1950** at the outer intermediate surface **1903**. The annular shoulder **1951** is symmetric; that is, the height, or longitudinal distance from the linear section **1966**, is equal at equal inward or outward radial distances from the crown **1992**. The symmetric splined curve profile of the annular shoulder **1951** defines the contact progression surface, and the linear section **1966** of the stress relief cutout **1952** serves as the reference surface for longitudinal height measurements.

[0380] The symmetric curve profile of the annular shoulder **1951** is defined by nine reference points: the innermost point **19107**, the crown **1992**, the outermost point **19109**, first intermediate point **19111**, second intermediate point **19113**, third intermediate point **19115**, fourth intermediate point **19117**, fifth intermediate point **19119**, and sixth intermediate point **19121**. The intermediate points **19111**, **19113**, and **19115** are spaced at regular radial intervals between the crown **1992** and the innermost point **19107**, and the intermediate points **19117**, **19119**, and **19121** are spaced at regular radial intervals between the crown **1992** and the outermost point **19109**. The crown **1992** is radially centered on the annular shoulder **1951** and located at the maximum height **19105** from the linear section **1966**. The innermost point **19107** has a height of 0.000000 inches relative to the linear section **1966** and is radially positioned where the annular shoulder **1951** would have intersected the linear section **1966** but for the presence of the eighteenth radius **1965**. The outermost point **19109** also has a height of 0.000000 inches relative to the linear section **1966** and is radially positioned at the outer intermediate surface **1903**, although it is a virtual point, as the curve is truncated at that location by the twelfth radius **1950**. The longitudinal height of each of the nine reference points relative to the linear section **1966** may be used to define the curve profile of the annular shoulder **1951** and may be modified as needed to achieve a desired contact progression.
[0381] In one specific implementation of the embodiment shown in FIGS. **173** and **174**, the annular shoulder **1951** has a total radial width of 0.302365 inches. The crown **1992** is radially centered and located at a radial position of 0.1511825 inches, with a height of 0.000700 inches relative to the linear section **1966**. The innermost point **19107** is located 0.151182 inches radially inward from the crown **1992** and has a height of 0.000000 inches. The outermost point **19109** is located 0.151182 inches radially outward from the crown **1992** and likewise has a height of 0.000000 inches. The first intermediate point **19111** is positioned 0.037796 inches radially inward from the crown **1992** and has a height of 0.000525 inches. The second intermediate point **19113** is positioned 0.075591 inches radially inward from the crown **1992** and has a height of 0.000350 inches. The third intermediate point **19115** is positioned 0.113387 inches radially inward from the crown **1992** and has a height of 0.000175 inches. The fourth intermediate point **19117** is positioned 0.037796 inches radially outward from the crown **1992** and has a height of 0.000525 inches. The fifth intermediate point **19119** is positioned 0.075591 inches radially outward from the crown **1992** and has a height of 0.000350 inches. The sixth intermediate point **19121** is positioned 0.113387 inches radially outward from the crown **1992** and has a height of 0.000175 inches. The longitudinal height values at each of these nine reference points define the curve profile of the annular shoulder **1951** and may be used to precisely control the shape of the contact progression surface formed between the fluid routing plug **1900** and the insert **19210**.
[0382] The third of the three embodiments, shown in FIGS. **175-176**, has a fluid routing plug **2000** with an annular shoulder **2051**. The annular shoulder **2051** comprises a single asymmetric splined curve having a crown **2092** at the point the annular shoulder **2051** contacts the front surface **20416** of the insert **20210**. The crown **2092** is radially centered on the annular shoulder **2051**. The crown **2092** is spaced longitudinally from the linear section **2066** of the stress relief cutout **2052**, and that spacing defines the height **20105** of the crown **2092**. The annular shoulder **2051** transitions to the linear section **2066** with an eighteenth radius **2065** and is truncated by the twelfth radius **2050** at the outer intermediate surface **2003**. The asymmetric nature of the annular shoulder **2051** arises from the differing longitudinal heights of the innermost and outermost points relative to the linear section **2066**. Specifically, the innermost point **20107** lies at a different longitudinal height than the outermost point **20109**, and the intermediate points between the crown and these two locations reflect that asymmetry. The asymmetric splined curve profile of the annular shoulder **2051** defines the contact progression surface, and the linear section **2066** of the stress relief cutout **2052** serves as the reference surface for longitudinal height measurements.
[0383] The asymmetric curve profile of the annular shoulder **2051** is defined by nine reference points: the innermost point **20107**, the crown **2092**, the outermost point **20109**, first intermediate

point **20111**, second intermediate point **20113**, third intermediate point **20115**, fourth intermediate point **20117**, fifth intermediate point **20119**, and sixth intermediate point **20121**. The intermediate points **20111**, **20113**, and **20115** are spaced at regular radial intervals between the crown **2092** and the innermost point **20107**, and the intermediate points **20117**, **20119**, and **20121** are spaced at regular radial intervals between the crown **2092** and the outermost point **20109**. The crown **2092** is located at the maximum height **20105** from the linear section **2066**. The innermost point **20107** is radially positioned where the annular shoulder **2051** would have intersected the linear section **2066** but for the presence of the eighteenth radius **2065** and has a height of 0.000000 inches relative to the linear section **2066**. The outermost point **20109** is radially positioned at the outer intermediate surface **2003**, although it is a virtual point due to truncation by the twelfth radius **2050**, and it has a nonzero height relative to the linear section **2066**. The differing heights of the innermost and outermost points **20107** and **20109** result in a curve that is asymmetrical in profile while remaining radially centered. The longitudinal height of each of the nine reference points relative to the linear section **2066** may be used to define the curve profile of the annular shoulder **2051** and may be modified as needed to achieve a desired contact progression.

[0384] In one specific implementation of the embodiment shown in FIGS. **175** and **176**, the annular shoulder **2051** has a total radial width of 0.302365 inches. The crown **2092** is radially centered and located at a radial position of 0.151182 inches, with a height of 0.000800 inches relative to the linear section **2066**. The innermost point **20107** is located 0.151182 inches radially inward from the crown **2092** and has a height of 0.000000 inches. The outermost point **20109** is located 0.151182 inches radially outward from the crown **2092** and has a height of 0.000500 inches. The first intermediate point **20111** is positioned 0.037796 inches radially inward from the crown **2092** and has a height of 0.000600 inches. The second intermediate point **20113** is positioned 0.075591 inches radially inward from the crown **2092** and has a height of 0.000400 inches. The third intermediate point **20115** is positioned 0.113387 inches radially inward from the crown **2092** and has a height of 0.000200 inches. The fourth intermediate point **20117** is positioned 0.037796 inches radially outward from the crown **2092** and has a height of 0.000725 inches. The fifth intermediate point **20119** is positioned 0.075591 inches radially outward from the crown **2092** and has a height of 0.000650 inches. The sixth intermediate point **20121** is positioned 0.113387 inches radially outward from the crown **2092** and has a height of 0.000575 inches. The longitudinal height values at each of these nine reference points define the curve profile of the annular shoulder **2051** and may be used to precisely control the shape of the contact progression surface formed between the fluid routing plug **2000** and the insert **20210**.

[0385] The embodiments described above illustrate various implementations of an annular shoulder having a contoured contact progression surface formed between a fluid routing plug and an insert. However, numerous alternative embodiments may be employed within the scope of the invention. For example, the asymmetric height profile of the contact progression surface, such as that described in the third embodiment, may be reversed so that the high side is positioned radially inward rather than outward. Likewise, the annular shoulder may have a symmetric radial geometry while maintaining an asymmetric longitudinal height profile, or it may have an asymmetric radial profile with a symmetric height.

[0386] The embodiments described above illustrate various implementations of an annular shoulder having a contoured contact progression surface formed between a fluid routing plug and an insert. However, numerous alternative embodiments may be employed within the scope of the invention. For example, the asymmetric height profile of the contact progression surface, such as that described in the third embodiment, may be reversed so that the high side is positioned radially inward rather than outward. Likewise, the annular shoulder may have a symmetric radial geometry while maintaining an asymmetric longitudinal height profile, or it may have an asymmetric radial profile with a symmetric height.

[0387] While the crown is shown in the figures as radially centered on the annular shoulder, this is not a requirement. The crown may be located at any radial position along the annular shoulder. Likewise, although the desired initial contact location is between the crown of the annular shoulder and the radial center of the front surface of the insert, the illustrations depict the initial contact point on the insert front surface as offset from the radial center. In alternative embodiments, the initial contact point on the insert may also be located at any radial position. The annular shoulder and the front surface of the insert may each be shaped or contoured to achieve a desired progression of contact during assembly, regardless of radial symmetry.

[0388] In each of the described embodiments, the contoured surface is formed on the annular shoulder of the plug, with the insert having a corresponding planar surface. In an alternative embodiment, the contoured surface may instead be formed on the insert, while the plug has a generally planar or complementary surface. The described geometry, reference points, and curve profiles may be applied to the insert rather than the plug.

[0389] The embodiments described herein generally depict an interface that is perpendicular to the longitudinal axis of the plug. However, in alternative configurations where the contact interface is angled relative to the longitudinal axis, such as those previously described with deformed mating surfaces, the contour of the annular shoulder may be adjusted accordingly. The reference heights of the various points on the contact progression surface may be projected or defined relative to a plane corresponding to the deformed surface, or compensated for in the design of the contour.

[0390] Additional variations may include modifying the rate of change of the contour height between the crown and the inner or outer reference points to produce a more gradual or more abrupt contact progression. Different radii of curvature or spline tension values may be employed to alter the contact behavior. The annular shoulder may also include locally flat or relief regions for stress control or manufacturing ease. Further, the overall diameter of the annular shoulder or the radial width of the contoured region may be adjusted without departing from the invention.

[0391] In all embodiments, the number of reference points used to define the contour may be increased or decreased, and the spacing between those points need not be uniform. Curves may be defined with as few as three points or more than nine, depending on the desired resolution and complexity of the surface.

[0392] In alternative embodiments, features of one of the various fluid routing plugs and corresponding housings described in the '529 application, previously incorporated herein by reference, may be included in one or more of the various fluid routing plugs and/or housings described herein. One of skill in the art will appreciate that the various housing and components described herein may have different shapes and sizes, depending on the shape and size of the various components chosen to assemble each fluid end section.

[0393] One or more kits may be useful in assembling a fluid end assembly out of the various fluid end sections described herein. A single kit may comprise a plurality of one of the various embodiments of housings and fasteners described herein. The kit may further comprise a plurality of one or more of the various inner components described herein. The kit may even further comprise a plurality of one or more of the various components attached to the various housings described herein.

[0394] The various features and alternative details of construction of the apparatuses described herein for the practice of the present technology will readily occur to the skilled artisan in view of the foregoing discussion, and it is to be understood that even though numerous characteristics and advantages of various embodiments of the present technology have been set forth in the foregoing description, together with details of the structure and function of various embodiments of the technology, this detailed description is illustrative only, and changes may be made in detail, especially in matters of structure and arrangements of parts within the principles of the present technology to the full extent indicated by the broad general meaning of the terms in which the appended claims are expressed.

Claims

1. A fluid end, comprising: a housing having a bore formed therein, the bore comprising a counterbore with a base surface; a fluid routing plug at least partially disposed within the bore, the fluid routing plug comprising: a body extending along a longitudinal axis having: a first end; a second end; and an outer surface extending between the first end and the second end, the outer surface defining an annular shoulder disposed proximate the second end, wherein: a first plurality of fluid passages extending from the outer surface to the second end; and a second plurality of fluid passages extending from the second end to the first end; and a hardened insert positioned between the annular shoulder of the fluid routing plug and the base surface of the counterbore and comprising: a pre-formed first surface having a first profile abutting the shoulder; a pre-formed second surface having a second profile abutting the base surface; wherein the annular shoulder is characterized by a contoured contact progression profile having an initial central contact at which the pre-formed first surface contacts the annular shoulder, wherein, as compressive load is applied to the hardened insert, the pre-formed first surface plastically deforms against the annular shoulder to increase the contact area radially outward and radially inward from the initial central contact.
2. The fluid end of claim 1 wherein the annular shoulder comprises a symmetric splined curve, wherein the initial central contact is a crown centered on the shoulder.
3. The fluid end of claim 1, wherein the annular shoulder comprises an asymmetric splined curve, wherein the initial central contact is a crown radially centered and a longitudinal height that varies unequally on opposite sides of the crown.
4. The fluid end of claim 1 wherein the annular shoulder comprises a flat central region flanked by tapered surfaces.
5. The fluid end of claim 1 wherein the fluid routing plug comprises a plurality of support columns positioned within the first plurality of fluid passages.
6. The fluid end of claim 1 wherein the base surface of the counterbore is shaped to match a post-deformation profile of the second surface of the insert.
7. The fluid end of claim 1, wherein the contoured contact progression profile of the annular shoulder comprises a symmetric splined curve, wherein the initial central contact is a crown located at a radial midpoint of the shoulder, the crown having a maximum height relative to a reference surface, and a plurality of radially spaced reference points positioned symmetrically on opposite sides of the crown, each reference point having a longitudinal height that decreases with increasing radial distance from the crown, terminating at an outermost point and an innermost point, each located on the reference surface.
8. The fluid end of claim 7 in which the crown has a distance from the reference surface of less than one one-thousandth of an inch.
9. The fluid end of claim 8 in which the crown is radially disposed less than one-quarter of an inch from the innermost point of the annular shoulder.
10. The fluid end of claim 1, wherein the contoured contact progression profile of the annular shoulder comprises an asymmetric splined curve, wherein the initial central contact is a crown located at a radial midpoint of the shoulder, the crown having a maximum height relative to a reference surface, and a plurality of radially spaced reference points on opposite sides of the crown, the reference points having longitudinal heights that decrease in magnitude with increasing radial distance from the crown, wherein the longitudinal height of the outermost point differs from the longitudinal height of the innermost point, the innermost point being located on the reference surface.
11. The fluid end of claim 10 wherein the crown has a distance from the reference surface of less than one-one thousandth of an inch, and a radial distance of less than one quarter of an inch from the innermost point of the annular shoulder.

12. The fluid end of claim 10 in which the reference points are spaced at regular radial intervals and wherein the angle of the annular shoulder relative to the reference plane changes at each reference point.

13. The fluid end of claim 1 wherein the outer surface of the fluid routing plug further comprises an annular groove configured to receive a seal.

14. The fluid end of claim 13 in which the seal, situated in the annular groove, abuts an inner surface of the hardened insert.

15. A contact progression surface formed on an annular shoulder of a fluid routing plug, the contact progression surface comprising: a crown positioned at a radial midpoint of the shoulder and having a maximum height relative to a reference surface; a first set of reference points located at regular radial intervals inward from the crown, each having a longitudinal height less than that of the crown; and a second set of reference points located at regular radial intervals outward from the crown, each having a longitudinal height less than that of the crown; wherein the longitudinal height of the contact progression surface relative to the reference surface decreases with increasing radial distance from the crown, such that contact between the annular shoulder and a mating insert surface expands radially inward and outward from the crown as compressive load is applied.

16. The contact progression surface of claim 15 wherein the longitudinal heights of the first and second set of reference points are symmetric about the crown, such that the contact progression surface forms a symmetric splined curve centered at the crown and the innermost and outermost reference points are located on the reference surface.

17. The contact progression surface of claim 15, wherein the longitudinal height of the outermost reference point relative to the reference surface differs from the longitudinal height of the innermost reference point, such that the contact progression surface forms an asymmetric splined curve.

18. The contact progression surface of claim 15, wherein the crown comprises a flat central region extending over a finite radial width, the flat central region flanked by a first tapered surface extending radially inward and a second tapered surface extending radially outward, each decreasing in longitudinal distance from the reference surface with increasing radial distance from the flat region.

19. A fluid end, comprising: a housing having a counterbore with a base surface; a fluid routing plug contained in the housing, the fluid routing plug having the contact progression surface of claim 15; a hardened insert, positioned between the contact progression surface of the fluid routing plug and the base surface of the counterbore.

20. The fluid end of claim 19, wherein the hardened insert comprises a pre-formed surface abutting the contact progression surface, the pre-formed surface configured to plastically deform under compressive load to increase contact area with the contact progression surface.

21. The fluid end of claim 20, wherein the pre-formed surface of the hardened insert has a concave profile prior to deformation and is configured to deform into a planar surface upon installation.